Kron-Reduced vs. Full Graph Laplacians for Water-Quality Regression on the Colorado River Network

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November 2026

Abstract

The graph-Laplacian water-quality models for the Colorado basin use a Kron (Schur-complement) reduction of the 67,401-node river network onto the 26 nodes with water-quality (WQ) stations. We test whether this reduction changes the model. For five MERIT-based edge weightings, on both the uncompressed and the compressed (67-node) graph, the Kron and full Laplacians agree to within 10^{-8} on every quantity the reduction should preserve. Re-fitting the nested leave-one-node-out (LOO) regression with the full Laplacian changes R^2 by at most 10^{-10} and selects identical hyperparameters in every fold. Kron reduction is therefore exact, and about $150\times$ faster. We also correct several problems in the original analysis. The most consequential is that feature sets chosen by looking at outer-LOO results overstated performance: by about 0.13 in R^2 for specific conductance and 0.02–0.08 for DO, pH and turbidity. With selection done inside the inner loop, the graph term clearly helps for pH and specific conductance, while seasonal and MERIT predictors matter much more for water temperature, DO and turbidity. In a full rerun of the eight-model benchmark on identical folds, the graph model ranks first for specific conductance and pH, within 0.005 of the best for water temperature and DO, and seventh of eight for turbidity. When every model also selects its features inside the inner loop, the graph model still wins clearly for specific conductance (0.207) and pH (0.130). It ties for water temperature, falls to fifth for DO and rises to 0.106 for turbidity, where gradient boosting leads with 0.391. Finally, removing edge weights changes almost nothing on the original graph. On the compressed graph, however, an unweighted Laplacian (one step per river reach), chosen inside the inner loop, raises the graph model's specific conductance R^2 from 0.207 to 0.385.

1 Setup

Data. Hourly USGS measurements at 28 stations, mapped to 26 river-network nodes. Depending on the target, 17–24 nodes have usable data (DO 17, water temperature 24, pH 19, specific conductance 24, turbidity 17). The network is the largest connected component of the MERIT-derived Colorado graph: 67,401 nodes and 67,400 edges, each edge about 70–120 m long.

Model. For node i and hour t, the prediction is $\hat{y}_{it} = \beta_0 + z_{it}^\top \beta + g_i$, where z_{it} are standardized predictors and g_i is a node offset. The fit minimizes

$$\sum_{i \in \text{train}} \frac{1}{n_i} \sum_t \left(y_{it} - \beta_0 - z_{it}^\top \beta - g_i\right)^2 \;+\; \lambda_\beta \sum_{k \geq 1} \beta_k^2 \;+\; \lambda\, g^\top \tilde{L}\, g, \qquad \text{s.t. } \textstyle\sum_{i \in \text{WQ}} g_i = 0, \tag{1}$$

with $g^\top L g = \sum_{(i,j)} w_{ij}(g_i - g_j)^2$ and $\tilde{L} = L/s$. The scale s is the mean positive diagonal of the Kron matrix. The weight $1/n_i$ makes each station count equally. A held-out node has no data term,

so its offset is set by the smoothness penalty alone. The hyperparameters λ and λ_β (and, where stated, the edge weighting) are tuned by an inner LOO. Performance is the node-weighted R^2 of the outer-LOO predictions.

Edge weightings. $w_{ij} = \exp(\text{score}_{ij})/d_{ij}$, where d_{ij} is the edge length in km and the score mixes z-scored log upstream area, log width and log slope. There are five schemes: `distance_only`, `area`, `width`, `slope` and `area_width`.

Kron reduction. Partitioning nodes into WQ nodes B and the rest I,

$$L_{\text{kron}} = L_{BB} - L_{BI}L_{II}^{-1}L_{IB} \in \mathbb{R}^{26\times 26}. \tag{2}$$

The non-WQ nodes have no data term in (1), so minimizing over their g in closed form leaves exactly $g_B^\top L_{\text{kron}} g_B$. The two formulations should therefore be the same model. This report checks that numerically.

2 Are the Laplacians the same?

The Kron and full matrices have different sizes, so we compare what Kron reduction must preserve:

- **Effective resistance** between every pair of WQ nodes, computed on the full Laplacian by sparse solves and on L_{kron}.
- **Dirichlet energy:** for random WQ values x_B with harmonic extension x, $x^\top Lx$ should equal $x_B^\top L_{\text{kron}} x_B$.
- **Compression:** Kron(uncompressed) vs Kron(compressed), compared directly.

As a reference for what a *wrong* reduction looks like, we also compare against the sub-matrix L_{BB}, i.e. simply deleting the non-WQ nodes.

Table 1: Largest relative errors, uncompressed graph. All values are at rounding-error level, except the deliberately wrong L_{BB} reference.

Weighting	Eff. resistance	Energy	Kron unc. vs comp.	Naive L_{BB} (reference)
`distance_only`	1.9×10^{-9}	2.4×10^{-12}	6.7×10^{-13}	205
`area`	4.7×10^{-9}	1.5×10^{-12}	5.1×10^{-13}	368
`width`	4.1×10^{-10}	3.1×10^{-12}	6.4×10^{-13}	253
`slope`	7.3×10^{-9}	8.8×10^{-12}	2.6×10^{-12}	257
`area_width`	6.0×10^{-9}	2.6×10^{-12}	8.1×10^{-13}	359

On the compressed graph, the resistance and energy errors are below 1.5×10^{-12} for every weighting. Compression is also exact. Each compressed edge replaces a chain of resistors in series, and series resistances add, so the total resistance between stations is unchanged.

3 Does Kron change the regression?

The nested LOO was re-run with four Laplacians, using exactly the same solver code:

Variant	Laplacian in (1)	Size
`kron_uncompressed`	Kron of the 67k graph (original method)	26×26
`kron_compressed`	Kron of the 67-node graph	26×26
`full_uncompressed`	full Laplacian, g on every node	$67{,}401 \times 67{,}401$
`full_compressed`	full Laplacian, g on every node	67×67

The full variants never form a Schur complement. They solve the complete (β, g) system by block elimination with a sparse factorization. Each full Laplacian is divided by the same s as its Kron reduction, so λ has the same meaning in all four.

Table 2: Outer-LOO R^2, chemistry predictors only. "Graph, $1/d$" is `distance_only`. The last column is the largest change in R^2 between `kron_uncompressed` and any other variant, over both graph modes.

Target	Linear	Graph, $1/d$	Graph, tuned weighting	Max $\lvert\Delta R^2\rvert$
DO concentration	0.6208	0.6135	0.6065	3.4×10^{-11}
Water temperature	−0.0556	−0.0346	−0.0575	9.9×10^{-11}
pH	−0.0352	0.0742	0.0742	1.6×10^{-11}
Specific conductance	−0.0891	0.0907	0.1062	2.2×10^{-11}
Turbidity	0.0228	−0.0486	−0.0360	4.4×10^{-11}

- **Identical models.** All four variants chose the same (weighting, λ, λ_β) in 100% of outer folds for every target. R^2 differs by at most 10^{-10}.
- **Reproduction.** `kron_uncompressed` reproduces the original notebook's saved results (`full_graph_nested_loo_results.csv` and the `Linear + Laplacian` rows of `all_models_node_loo_results.csv`) to within 1.3×10^{-12}.
- **Cost.** Kron took 0.1 s per outer fold vs 15.2 s for the full 67k Laplacian (26 minutes in total), about 150× faster.

When the graph term helps. With chemistry predictors only, the graph term improves pH (+0.11) and specific conductance (+0.18 to +0.20). Both are plausibly controlled by geology and therefore spatially structured. It is neutral or slightly harmful for DO, water temperature and turbidity. Tuning the MERIT weighting instead of using $1/d$ changes R^2 by at most 0.02. The weighting chosen is also unstable from fold to fold for DO (`width` in 13 of 17 folds) and for water temperature (`area` 15, `slope` 5, `area_width` 4 of 24). For pH, `distance_only` was chosen in all 19 folds.

4 Corrections to the original analysis

4.1 pH transforms with the scaled Laplacian

Predictors are specific conductance plus $K = 1$ Fourier terms.

Raw and z-scored pH give identical results, as expected for an affine transform. Fitting H^+ is worse, and no H^+ predictions had to be clipped. Raw pH should be kept.

Table 3: Problems found in `largest_uncompressed_laplacian.ipynb` and how they were handled.

Problem	Fix
pH-transform section used the *unscaled* Kron Laplacian (λ about $21\times$ weaker)	scaled Laplacian; unscaled run alongside for comparison
Static feature sets and Fourier K chosen from outer-LOO results, then re-scored on the same folds	selection moved inside the inner LOO
"Discharge" edge feature is a linear function of log UPA, so it duplicates `area`; its fit also used held-out stations	discharge removed
HAND is 0 at all 26 WQ nodes (river pixels)	HAND removed from the predictors
Comparison tables used hard-coded R^2 values from earlier runs	all tables computed from the run

Table 4: pH nested-LOO R^2. The scaling fix changes graph R^2 by only -0.002, because tuning λ over a wide grid absorbs the constant factor.

Target transform	Linear	Graph, scaled (fixed)	Graph, unscaled (original)	Change
raw pH	0.0163	0.1300	0.1315	-0.0016
z-scored pH	0.0163	0.1300	0.1315	-0.0016
$H^+ = 10^{7-\mathrm{pH}}$	-0.0662	0.0930	0.0954	-0.0024

4.2 Feature selection inside the nested LOO

In each outer fold, the feature set was chosen by inner-LOO MSE alone, separately for the linear and graph models (graph: Kron Laplacians, with weighting, λ and λ_β tuned). The 30 candidates are chemistry plus Fourier $K \in \{1, \ldots, 5\}$, combined with one of six static options: none, one of elevation, log UPA, log width or local slope, or all four. The same run gives each candidate's own nested-LOO R^2. Taking the best of those reproduces what the original analysis effectively reported, so the difference measures how much it was overstated.

- **Water temperature.** A large and genuine gain, from 0.68 to 0.86, with almost no overstatement. All four MERIT features were chosen in every fold. With them included, the graph term adds nothing, because the node-level features already explain the offsets the Laplacian was capturing.
- **DO.** A modest genuine gain, to 0.70–0.71, driven by $K = 2$–4 seasonal terms with log UPA (graph, 13 of 17 folds) or elevation (linear, 14 of 17 folds). The graph is slightly worse than linear.
- **Specific conductance.** The largest overstatement. Choosing local slope after seeing outer results inflated graph R^2 from a fair 0.14 to 0.27, and the fair value is no better than the plain $K = 1$ baseline (0.146). The graph term is still the only model with positive R^2. Local slope was chosen in 22 of 24 folds, so it may be real, but it does not improve held-out accuracy.
- **pH.** The graph term helps (0.10 vs -0.06 for linear), but R^2 stays low and the chosen features vary from fold to fold.

Table 5: Outer-LOO R^2 with seasonal and MERIT predictors. "Biased" is the best candidate chosen after seeing outer results. "Fair" is the feature set chosen inside the inner loop, and is the number to report. "Baseline" is $K = 1$ with no static features.

Target	Model	Baseline	Biased	**Fair**	Overstated
Water temperature	linear	0.6781	0.8599	0.8573	0.0026
	graph	0.7394	0.8591	0.8568	0.0024
DO concentration	linear	0.6908	0.7260	0.7109	0.0151
	graph	0.6831	0.7220	0.7009	0.0211
pH	linear	0.0163	0.0268	−0.0553	0.0821
	graph	0.1300	0.1439	0.0977	0.0461
Specific conductance	linear	−0.0793	−0.0564	−0.1690	0.1126
	graph	0.1462	0.2674	0.1403	0.1271
Turbidity	linear	0.0554	0.1954	0.1443	0.0511
	graph	−0.0084	0.1064	0.0740	0.0324

- **Turbidity.** Linear with $K = 5$ seasonal terms and no static features is best (0.14). The graph term hurts.

5 Reconciliation with the Oct 29 and Nov 5 meeting notes

The model and theory in the meeting notes agree with this report. That includes the objective (1), the $1/n_i$ weighting, the Kron/Schur derivation, preservation of effective resistance, and the compression result. The reported Graph-Laplacian R^2 values differ. Every one of them was copied from an output cell of `largest_uncompressed_laplacian.ipynb`. Table 6 traces each one and gives two corrected values:

- **Benchmark refit:** the graph model fitted inside the updated `water_quality_model_benchmark.py`, on exactly the same rows, predictors and folds as the conventional models.
- **Fair:** the value from Table 5, with features selected inside the inner loop.

Table 6: Graph-Laplacian R^2 as reported in the notes, where each value came from, and the corrected values.

Target	Notes	Source in original notebook	Benchmark refit	Fair
Water temperature	0.8628	cell 118: reduced static set	0.8628	0.8568
DO concentration	0.7261	cell 131: DO base set, K tuned	0.7306	0.7009
Specific conductance	0.2525	cell 143: SC base set, K_{year} tuned	0.2569	0.1403
pH	0.1316	cell 67: unscaled Laplacian	0.1300	0.0977
Turbidity	0.0145	cell 129: other features/rows	0.0264	0.0740

Why the numbers differ.

1. **Post-hoc feature selection.** The Oct 29 notes state that "the outer held-out node is never used for tuning" (§9.1). That holds for λ, λ_β and K, but not for the static feature sets (`REDUCED_STATIC`, `DO_BASE_FEATURES`, `COND_BASE_FEATURES`). These were chosen by comparing

outer-LOO R^2 across candidate sets. This accounts for most of the gap between the notes and the fair values, above all for specific conductance ($0.25 \rightarrow 0.14$).

2. **The Nov 5 graph rows were not produced by the benchmark.** The original `water_quality_model_benchmark.py` never fitted the graph model. It read its rows from a `graph_laplacian_metrics.csv`, which does not exist, so the values in the notes were entered by hand.

3. **Turbidity was compared on different data.** The graph row was labelled 90,843 rows, but its value (0.0145, RMSE 561.81) comes from a model using specific conductance + pH + K=1 + slope on 88,095 rows. The conventional turbidity models use only slope + K=1 (`TURB_FEATURES`). On those rows and predictors, the graph model scores 0.026.

4. **Fixed vs tuned K.** For DO and specific conductance, the notes' graph values tuned K inside the loop, while the conventional models used fixed K (K=2; K_{day}=1, K_{year}=4). Refitting the graph model with the same fixed K changes R^2 by less than 0.005.

5. **pH scaling.** The notes' 0.1316 used the unscaled Laplacian. The scaled value is 0.1300, a negligible difference.

Updated benchmark. `water_quality_model_benchmark.py` (and the matching notebook) now fits the Graph-Laplacian model inside the same nested LOO as the seven conventional models. It uses the five saved Kron Laplacians, with weighting, λ and λ_β tuned by the same inner-LOO criterion. Fitting the graph model takes about 20 s for all five targets.

Full benchmark rerun. The complete benchmark (all eight models, all five targets) was rerun on the lab server with the updated script. The results are saved in `results/benchmark_results/`. Table 7 gives every model's R^2. Table 8 compares the graph model with the best conventional model and with the value in the notes. The graph values from the full run match the graph-only run exactly. The conventional models' values are unchanged from the Nov 5 table.

Table 7: Full benchmark rerun: node-weighted nested-LOO R^2 for every model on identical rows, predictors and folds. Bold marks the best model for each target. Stations and rows: water temperature 24 / 144,980; DO 17 / 114,607; specific conductance 24 / 144,980; pH 19 / 121,011; turbidity 17 / 90,843.

Model	Water temp.	DO	Spec. cond.	pH	Turbidity
Graph-Laplacian	0.8628	0.7306	**0.2569**	**0.1300**	0.0264
OLS	0.8634	0.7305	−0.1242	0.0116	0.0476
Ridge	**0.8640**	0.7252	−0.0997	0.0130	0.0393
Elastic Net	0.8635	0.7002	−0.1163	0.0226	0.0360
Spline GAM	0.8497	0.6458	0.0307	0.0023	0.1023
Random Forest	0.7541	0.6933	−0.0852	−0.0231	0.1003
Gradient Boosting	0.8221	**0.7358**	0.0011	0.0035	**0.3069**
Dummy	−0.0327	−0.0286	−0.0623	−0.0301	−0.0271
Graph rank (of 8)	4	2	1	1	7

The rankings in the Nov 5 notes are unchanged. The graph model is best for specific conductance (by 0.23 over the next model) and pH (by 0.11). It is in a near tie with the linear models for water temperature: four models fall within 0.0012 of each other, well inside the noise. It is second for DO,

Table 8: Best conventional model vs the Graph-Laplacian model from the full rerun, and the graph value reported in the notes. The feature sets are those in `TARGET_CONFIGS`, which were chosen post hoc, so every model's R^2 here is somewhat optimistic.

Target	Best conventional	R^2	Graph-Laplacian	Notes (graph)
Water temperature	Ridge	0.8640	0.8628	0.8628
DO concentration	Gradient Boosting	0.7358	0.7306	0.7261
Specific conductance	Spline GAM	0.0307	0.2569	0.2525
pH	Elastic Net	0.0226	0.1300	0.1316
Turbidity	Gradient Boosting	0.3069	0.0264	0.0145

0.005 behind gradient boosting and level with OLS. For turbidity it ranks seventh, ahead only of the dummy model, while gradient boosting reaches 0.31. Turbidity seems to depend on nonlinear effects that neither the linear models nor the graph model capture. The RMSE ranking agrees: the graph model has the lowest RMSE for specific conductance (382 vs at least 437 µS/cm) and for pH (0.221 vs at least 0.234). What changes is the size of the advantage once feature selection is done fairly (Table 5). For specific conductance, the graph model's fair R^2 is about 0.14, not 0.25. The conventional models also need their feature sets selected inside the loop before their numbers can be compared fairly. Section 6 does this.

6 Benchmark with inner-loop feature selection

Method. `water_quality_model_benchmark.py` now selects each model's feature set inside the nested LOO (the default, `BENCHMARK_FEATURE_SELECTION=inner`). In every outer fold, every model chooses its features using inner-LOO MSE on the training stations only. The held-out station never influences the choice.

- **Candidates.** The chemistry predictors are fixed per target. They are the only columns with missing values, so every candidate uses exactly the same rows as before. The seasonal and static parts vary: Fourier $K \in \{1, 2, 4\}$ × static ∈ {none, `local_slope`, elevation + log area + slope}. The `TARGET_CONFIGS` set is added as a further candidate ("configured"), giving 9–10 candidates per target.
- **Search.** The graph model searches feature set, weighting, λ and λ_β jointly. The other models use two stages: they choose the feature set at the middle setting of their grid, then tune their settings on that set. The dummy model ignores its inputs and is not affected.
- **Run.** One full server run, with no row thinning (`BENCHMARK_INNER_STRIDE=1`), on the same stations, rows and folds as Table 7.

Results.

- **Specific conductance.** The graph model drops by 0.05, to 0.207. This confirms that the fixed-feature 0.257 was inflated. It is still the only model clearly above zero, ahead of the next one (spline GAM, 0.022) by about 0.19.
- **pH.** The graph model is unchanged at 0.130 and still leads by about 0.11. The tree models collapse (gradient boosting −0.17, random forest −0.27). Given richer seasonal and static candidates, they choose sets that fit the training stations well but transfer poorly to a new station.

Table 9: Benchmark with inner-loop feature selection: node-weighted nested-LOO R^2. Rows, stations and folds are identical to Table 7. Bold marks the best model for each target.

Model	Water temp.	DO	Spec. cond.	pH	Turbidity
Graph-Laplacian	0.8628	0.6998	**0.2068**	**0.1299**	0.1063
OLS	0.8634	0.6951	−0.0987	−0.0138	0.1360
Ridge	**0.8640**	0.7008	−0.1096	0.0029	0.1362
Elastic Net	0.8635	0.7048	−0.1089	0.0226	0.1362
Spline GAM	0.8382	0.7076	0.0218	0.0117	0.0910
Random Forest	0.7579	0.6845	−0.1740	−0.2685	0.0996
Gradient Boosting	0.8326	**0.7327**	−0.0939	−0.1713	**0.3911**
Dummy	−0.0327	−0.0286	−0.0623	−0.0301	−0.0271
Graph rank (of 8)	4	5	1	1	5

Table 10: Change in R^2 from fixed features (Table 7) to inner-loop selection (Table 9). Negative values mean the fixed feature set had been flattering the model, or that inner selection picked worse features in some folds.

Model	Water temp.	DO	Spec. cond.	pH	Turbidity
Graph-Laplacian	0.000	−0.031	−0.050	0.000	+0.080
OLS	0.000	−0.035	+0.025	−0.025	+0.088
Ridge	0.000	−0.024	−0.010	−0.010	+0.097
Elastic Net	0.000	+0.005	+0.007	0.000	+0.100
Spline GAM	−0.011	+0.062	−0.009	+0.009	−0.011
Random Forest	+0.004	−0.009	−0.089	−0.245	−0.001
Gradient Boosting	+0.011	−0.003	−0.095	−0.175	+0.084

- **Water temperature.** Nothing material changes. The four linear-type models, including the graph model, remain within 0.0012 of each other.
- **DO.** The graph model loses 0.031 and falls from second to fifth. Its hand-picked set suited it particularly well. The top five models now fall within 0.033 of each other, which is within noise for 17 stations.
- **Turbidity.** Almost every model improves. The graph model goes from 0.026 to 0.106, the linear models from about 0.04 to 0.136, and gradient boosting from 0.307 to 0.391. The hand-picked turbidity set (Fourier $K = 1$ plus slope) was a poor choice, and the inner loop finds better ones. Gradient boosting's large lead suggests nonlinear effects that the additive models miss.

The headline result survives fair selection: the graph model wins clearly for specific conductance and pH, and for no other target.

Relation to the fair values in Table 5. For specific conductance, the benchmark's 0.207 is higher than the 0.14 from the notebook's nested selection. The candidate sets differ. The benchmark includes the "configured" set (separate daily $K = 1$ and annual $K = 4$ terms plus slope), and the graph model picks it in 22 of 24 outer folds. The notebook's 30 candidates do not include it. That set was itself designed by looking at outer-LOO results, so offering it as a candidate carries over some of the original optimism. The benchmark values are fair *between models*, since all eight models

see the same candidates. As an absolute estimate for specific conductance, 0.14 remains the more conservative figure.

7 Effect of edge weights

Every Laplacian so far has used edge weights: 1/km, or the MERIT-based schemes. Here the weights are removed ($w = 1$ on every edge) on both the original and the compressed graph. The script is `compare_edge_weights.py`. Both graphs are trees (67,401 nodes with 67,400 edges, and 67 nodes with 66 edges), so the effective resistance between two stations is just the sum of $1/w$ along the river path between them:

- **distance:** km of river (median over station pairs 844 km);
- **unweighted, original graph:** number of original segments (median 8,758);
- **unweighted, compressed graph:** number of compressed reaches (median 9, range 1–27).

A fourth Laplacian, *hops*, gives each compressed reach weight 1/(original segments in the reach), which is the unweighted original graph's segments in series.

The Laplacians. Table 11 compares the Kron matrices after the model's scaling. Since λ is tuned, only relative weights matter.

Table 11: Kron-reduced Laplacians with and without edge weights, compared after scaling. "Resistance corr." is the correlation of pairwise effective resistances between WQ stations.

Comparison	Rel. Frobenius diff.	Resistance corr.
distance: original vs compressed	5.7×10^{-13}	1.0000
unweighted: original vs compressed	0.78	0.7926
unweighted original vs hops compressed	2.1×10^{-13}	1.0000
original graph: distance vs unweighted	0.034	0.9999
compressed graph: distance vs unweighted	0.79	0.7871

- **On the original graph, weights barely matter.** The 67k segments are nearly all the same length (about 0.1 km), so counting segments is almost the same as measuring km.
- **Without weights, compression is no longer exact.** Each compressed reach counts as one step, whether it is 5 km or 245 km long. Compression stays exact only when a reach keeps the series weight of the segments it replaces. The hops Laplacian does this and matches the unweighted original graph to 10^{-13}, as the distance weights do.

Effect on R^2, chemistry predictors only. Table 12 reruns the nested LOO of Section 3 (same predictors, grids and folds) with each Laplacian. The distance columns reproduce Table 2.

On the original graph, dropping the weights changes R^2 by at most 0.002, with nearly the same λ chosen in every fold. On the compressed graph, it lowers pH, DO and turbidity by 0.01–0.02 but raises specific conductance from 0.09 to 0.24.

Selected fairly inside the benchmark. Table 12 picks the best of several Laplacians after seeing outer R^2, which is the optimism problem of Section 4.2. To test it fairly, `unweighted_compressed` was added as a sixth weighting the graph model can choose inside the inner loop of the

Table 12: Nested-LOO R^2 by edge weighting, chemistry predictors only. Hops on the compressed graph equals unweighted on the original graph to four decimals and is not shown.

Target	No graph	Distance	Unweighted, original	Unweighted, compressed
DO concentration	0.6208	0.6135	0.6131	0.5992
Water temperature	−0.0556	−0.0346	−0.0341	−0.0334
pH	−0.0352	0.0742	0.0732	0.0519
Specific conductance	−0.0891	0.0907	0.0893	0.2420
Turbidity	0.0228	−0.0486	−0.0494	−0.0633

benchmark, alongside feature set, λ and λ_β. With this weighting, λ chose the bottom of the grid (0.01) in most folds, so the grid was extended down to 10^{-4}. Table 13 gives graph-only runs with inner feature selection. The other seven models do not use the Laplacian, so their values in Table 9 are unchanged.

Table 13: Graph-Laplacian R^2 in the benchmark with inner-loop feature selection, before and after adding the unweighted compressed Laplacian as a candidate weighting, and the weighting chosen across outer folds. "Best other" is the best of the seven other models in Table 9.

Target	5 weightings	+ unweighted	Best other	Weighting chosen
Specific conductance	0.2068	0.3847	0.0218	unweighted, 24/24
pH	0.1299	0.1173	0.0226	distance, 15/19
Water temperature	0.8628	0.8634	0.8640	unweighted, 24/24
DO concentration	0.6998	0.6996	0.7327	mostly distance
Turbidity	0.1063	0.1043	0.3911	width, 14/17

- **Specific conductance.** The inner loop picks the unweighted compressed Laplacian in all 24 folds, with $\lambda = 0.03$ in 21 of them, well inside the grid. R^2 rises from 0.207 to 0.385, the same with the original or extended λ grid, so the whole gain comes from the weighting. The graph model now leads every other model by about 0.36.
- **pH.** The graph model drops by 0.013. Two folds choose the unweighted Laplacian and predict worse. With the original λ grid the value is 0.115, so the extended grid is not the cause. On 19 stations this is within noise.
- **Water temperature.** The unweighted Laplacian is chosen, but with $\lambda = 100$, very strong smoothing, so the graph term is close to a constant and R^2 barely moves.
- **DO and turbidity.** No material change.

Interpretation. For specific conductance, the number of reaches between two stations describes how they differ far better than the km of river between them. Compressed-graph nodes sit at confluences and stations, so a reach count is roughly a count of tributary junctions. That is consistent with conductance being driven by tributary inputs and mixing at confluences more than by distance downstream. It is a hypothesis this analysis cannot confirm.

8 Final model comparison

Table 14 puts the updated graph model, with six candidate weightings including `unweighted_compressed` and the extended λ grid, next to the other seven models from the inner-selection benchmark (Table 9). Every model chooses its feature set inside the inner loop, on identical stations, rows and folds. Both columns come from server runs.

Table 14: Final comparison: node-weighted nested-LOO R^2 with inner-loop feature selection for every model. The Graph-Laplacian row is the updated model of Table 13. The other rows repeat Table 9. Bold marks the best model for each target.

Model	Water temp.	DO	Spec. cond.	pH	Turbidity
Graph-Laplacian	0.8634	0.6996	**0.3847**	**0.1173**	0.1043
OLS	0.8634	0.6951	−0.0987	−0.0138	0.1360
Ridge	**0.8640**	0.7008	−0.1096	0.0029	0.1362
Elastic Net	0.8635	0.7048	−0.1089	0.0226	0.1362
Spline GAM	0.8382	0.7076	0.0218	0.0117	0.0910
Random Forest	0.7579	0.6845	−0.1740	−0.2685	0.0996
Gradient Boosting	0.8326	**0.7327**	−0.0939	−0.1713	**0.3911**
Dummy	−0.0327	−0.0286	−0.0623	−0.0301	−0.0271
Graph rank (of 8)	4	5	1	1	5
Graph − best other model	−0.001	−0.033	+0.363	+0.095	−0.287

- **Specific conductance.** The graph model is the only model that predicts unseen stations usefully. Every other model is at or below zero except the spline GAM (0.022). The graph model reaches 0.385, with RMSE 348 µS/cm against at least 439 for the others.
- **pH.** The graph model is the only model meaningfully above zero, ahead by 0.095. The lead is at the edge of what 19 stations can resolve, and pH R^2 is low for every model.
- **Water temperature.** The graph model is in a four-way tie with ridge, elastic net and OLS, all within 0.0012. Seasonality explains almost everything, and the graph term adds nothing.
- **DO.** The graph model is in the middle of a tight group. The top five models span 0.033, within noise for 17 stations, with gradient boosting slightly ahead.
- **Turbidity.** Gradient boosting leads by a wide margin (0.391 against 0.136 for the linear models and 0.104 for the graph model). This points to nonlinear responses that no additive model captures.

Both targets where the graph model wins vary mainly *between* stations, which is what the river-network term captures. The other models see only a station's own predictors and season, and have nothing that relates one station to another. Compared with the Nov 5 benchmark (Table 7), the graph model keeps first place for specific conductance and pH, with a much larger margin for specific conductance. It drops from second to mid-pack for DO, where its hand-picked features had favoured it, and rises from seventh to fifth for turbidity, where every model improved once features were chosen fairly.

9 Ablation and reverse ablation

Design. `ablation_study.py` measures how much each input contributes to the graph-Laplacian

model. The graph term itself counts as one of the "variables". For each target they are:

- the target's chemistry predictors (as in the benchmark, so rows never change);
- the daily cycle (`hour_sin/cos_k`) and the annual cycle (`doy_sin/cos_k`), $K = 4$ each;
- the four static MERIT variables, one at a time: elevation, log upstream area, log width and local slope (`hand_m` is 0 at every station and is left out);
- the graph term g.

Ablation removes one variable at a time from the *full* model, which has all of them. **Reverse ablation** adds one variable at a time to the *base* model, which is intercept only. Without the graph term the model is ridge regression on the same features. Every configuration is its own nested LOO on the benchmark's rows: the inner loop tunes λ_β and, when the graph term is on, the weighting (six, including `unweighted_compressed`) and λ (the extended grid). That makes 90 configurations, which run in under two minutes.

Table 15: Ablation: change in R^2 when one variable is removed from the full model. Large negative values mark variables whose contribution no other variable can replace. Positive values mark variables that hurt.

Variable removed	Water temp.	DO	Spec. cond.	pH	Turbidity
Full-model R^2	*0.8598*	*0.7215*	*0.3149*	*0.1314*	*0.0894*
Graph term	0.000	−0.008	**−0.449**	**−0.109**	+0.047
Annual cycle	**−0.721**	−0.010	−0.107	−0.035	**−0.146**
Daily cycle	−0.009	−0.116	+0.003	−0.043	+0.001
Water temperature	–	**−0.211**	+0.025	–	–
Specific conductance	−0.002	−0.004	–	−0.006	–
Elevation	−0.030	−0.032	−0.042	0.000	−0.003
Log upstream area	−0.006	−0.021	+0.078	0.000	+0.004
Log width	+0.002	+0.004	+0.030	+0.004	+0.006
Local slope	−0.039	−0.006	−0.150	−0.026	−0.004

Table 16: Reverse ablation: change in R^2 when one variable is added to the intercept-only base model. It measures how much each variable explains on its own.

Variable added	Water temp.	DO	Spec. cond.	pH	Turbidity
Base-model R^2	*−0.0327*	*−0.0286*	*−0.0623*	*−0.0301*	*−0.0271*
Graph term	+0.024	−0.107	**+0.326**	**+0.067**	−0.017
Annual cycle	**+0.728**	+0.552	+0.002	+0.023	**+0.177**
Daily cycle	+0.009	+0.051	0.000	+0.052	−0.002
Water temperature	–	**+0.645**	−0.027	–	–
Specific conductance	−0.023	−0.064	–	−0.005	–
Elevation	+0.111	+0.011	+0.009	−0.011	−0.015
Log upstream area	0.000	+0.053	−0.041	−0.030	−0.023
Log width	−0.041	−0.012	+0.009	+0.010	−0.006
Local slope	+0.150	+0.050	−0.046	−0.007	−0.005

Results.

- **The graph term is the key variable for specific conductance and pH.** It is the most valuable variable for both targets, whether added alone or removed from the full model. For specific conductance, removing it costs 0.45; only the annual cycle for water temperature matters more for any target. In the full model the inner loop picks the `unweighted_compressed` weighting in all 24 folds for specific conductance, and distance in 17 of 19 for pH.
- **Water temperature is seasonality.** The annual cycle alone gives $R^2 \approx 0.70$. Slope and elevation add a little, consistent with steep, high streams being colder. The graph term adds nothing.
- **DO follows water temperature.** Water temperature is the most valuable variable both ways, consistent with oxygen solubility falling as water warms. The daily cycle adds 0.12 on top of everything else, consistent with photosynthesis by day and respiration at night. The annual cycle is useful alone (+0.55) but redundant once water temperature is in the model.
- **Turbidity depends on the annual cycle only**, likely snowmelt and storm seasons. The graph term hurts it: removing it from the full model gains 0.047. This matches the graph model's weak turbidity results in the benchmark.
- **The static MERIT variables matter mainly through slope.** Slope helps water temperature and specific conductance. Elevation, log area and log width are small or harmful. Log width hurts every target in the full model, and log area costs specific conductance 0.078.

Reading the two tables together. The static variables and the graph term are all constant within a station, so they compete to explain the same between-station differences. A variable with a large reverse-ablation gain but a small ablation loss is useful but redundant (for example the annual cycle for DO, and slope for water temperature). The opposite also occurs. For specific conductance, slope alone *hurts* (−0.046), yet removing it from the full model costs 0.150, the second-largest loss for that target. Elevation shows the same pattern on a smaller scale (+0.009 alone, −0.042 when removed). So slope does not predict conductance by itself; it corrects the between-station pattern that the graph term captures. For DO, the graph term alone is worse than the intercept-only model (−0.107): smoothing DO along the network makes predictions at unseen stations worse. DO varies much more over time (with temperature, and between day and night) than between stations, so the graph term has little real spatial signal to work with. In the full model its effect is negligible (−0.008). Note also that the full model is not the best model: for specific conductance it scores 0.315, against 0.385 for the selected model in Table 14, because log area and log width hurt it. That is why the benchmark selects features rather than using all of them.

9.1 All chemistry variables as predictors

The main ablation uses only each target's fixed chemistry predictors, so that its rows match the benchmark. Run with `--all-chemistry`, `ablation_study.py` instead tests all four other water-quality variables as predictors, alongside the seasonal cycles, the MERIT variables and the graph term (11 variables per target, 120 configurations). A row can only be used if all five variables are measured, so every target is limited to the same **14 stations and 80,592 rows**. These R^2 values are therefore not comparable with the rest of the report. The run shows which chemistry variables carry information, not how well each target can be predicted.

- **pH and DO are tightly linked.** Each is the other's most valuable predictor once removed from the full model (DO for pH −0.109; pH for DO −0.094, ahead of water temperature at −0.091). DO alone raises pH R^2 by 0.188. This is consistent with photosynthesis: algae take

Table 17: All-chemistry ablation (14 stations): change in R^2 when one variable is removed from the full model. Bold marks the largest loss for each target.

Variable	Water temp.	DO	Spec. cond.	pH	Turbidity
Full-model R^2	*0.8857*	*0.7859*	*0.2352*	*0.2937*	*−0.0609*
Graph term	+0.005	−0.007	**−0.418**	−0.024	+0.022
Annual cycle	**−0.207**	−0.011	−0.151	−0.020	−0.012
Daily cycle	−0.025	−0.059	+0.003	+0.009	+0.006
Water temperature	–	−0.091	+0.022	+0.059	−0.004
DO concentration	−0.056	–	+0.046	**−0.109**	+0.045
Specific conductance	−0.002	−0.001	–	+0.058	+0.001
pH	+0.014	**−0.094**	+0.014	–	−0.071
Turbidity	+0.001	+0.015	+0.004	−0.002	–
Elevation	+0.004	−0.007	+0.004	−0.006	+0.037
Log upstream area	+0.007	+0.008	+0.004	+0.087	**−0.086**
Log width	−0.008	+0.026	−0.022	+0.112	+0.023
Local slope	+0.002	+0.011	−0.008	+0.096	+0.002

up CO_2 during the day, which raises pH and DO together. The main ablation cannot show this, because pH there has only specific conductance as a chemistry predictor.

- **With DO available, the graph term matters much less for pH** (−0.024 when removed, against −0.109 in Table 15). Part of what the graph term captured for pH may be between-station differences in biological activity, which DO measures directly.
- **For specific conductance the graph term remains essential** (−0.418 when removed), even with every chemistry variable available. This is the most robust result of either ablation. Alone it does not help on these 14 stations (−0.044): interpolating between stations needs a denser network.
- **DO alone predicts water temperature well** (+0.720), because both follow the seasons, but it adds only 0.056 once the annual cycle is in the model.
- **Turbidity cannot be predicted from these variables.** The full model scores below the intercept-only model (−0.061). With 11 variables and 14 stations it overfits, and no single variable adds more than 0.063.
- **The full models overfit.** For pH, dropping log width, slope or log area each improves R^2 by 0.09–0.11. With this few stations, "use everything" is clearly worse than a selected subset.

10 Further graph-model variants

Four changes to the graph model were fixed in advance and tested together. Each is a candidate the inner loop may choose, so the outer held-out station never influences the choice:

1. **Confluence count** (`confluence_compressed`). The 67 compressed nodes are 26 stations (all mid-reach), 18 confluences, 20 headwater or outlet ends and 3 other gauges, so the reach count of Section 7 also counts any stations and gauges in between. In this Laplacian the graph distance counts only confluences: each edge has resistance 0.5 per confluence at its ends plus 0.1, so a pass-through node adds only 0.1.

Table 18: All-chemistry reverse ablation (14 stations): change in R^2 when one variable is added to the intercept-only base model. Bold marks the largest gain for each target.

Variable	Water temp.	DO	Spec. cond.	pH	Turbidity
Base-model R^2	*−0.0385*	*−0.0263*	*−0.1014*	*−0.0508*	*−0.0134*
Graph term	+0.097	+0.030	−0.044	+0.066	+0.001
Annual cycle	**+0.810**	+0.593	**+0.146**	+0.038	**+0.063**
Daily cycle	+0.008	+0.043	0.000	+0.066	0.000
Water temperature	–	**+0.723**	+0.019	−0.004	+0.040
DO concentration	+0.720	–	+0.118	**+0.188**	+0.036
Specific conductance	+0.022	+0.121	–	−0.010	−0.004
pH	+0.001	+0.165	−0.005	–	+0.008
Turbidity	+0.046	+0.043	−0.003	+0.016	–
Elevation	+0.043	+0.062	−0.040	−0.067	−0.025
Log upstream area	−0.006	+0.077	−0.036	−0.069	−0.044
Log width	−0.041	−0.016	−0.125	+0.009	−0.028
Local slope	0.000	0.000	−0.005	−0.002	0.000

2. **Length-power weights** $w = \ell^{-\alpha}$ with $\alpha \in \{0.25, 0.5, 0.75\}$, between the reach count ($\alpha = 0$) and distance ($\alpha = 1$).
3. **Log target** for turbidity and specific conductance. The model is fitted on $\log(1 + y)$ and transformed back with a node-weighted smearing correction. Raw vs log is chosen by inner-LOO MSE on the original scale.
4. **Wider λ grid**, 10^{-6} to 3000, since earlier runs chose the edges of the grid.

This gives 10 weightings and 17 λ values. The run is graph-only with inner feature selection, on the same rows and folds as Table 13.

Table 19: Graph-Laplacian R^2 with the four variants as inner-loop candidates, against the six-weighting model of Table 13, and the settings the inner loop chose across outer folds.

Target	Six wtgs.	Variants	Change	Chosen across outer folds
Spec. conductance	0.3847	0.3510	−0.0337	confluence 13, reach 10; raw 14, log 10
pH	0.1173	0.1248	0.0075	$\alpha = 0.5$–0.75: 13/19; $\lambda = 10^{-6}$: 17/19
DO concentration	0.6996	0.7026	0.0030	confluence 13/17
Water temperature	0.8634	0.8635	0.0001	reach 24/24; $\lambda = 3000$ in 22/24
Turbidity	0.1043	0.1023	−0.0020	width 14/17; raw 17/17

Results. No variant improves the model. Every change is within the noise level for 17–24 stations, and specific conductance falls by 0.034, at the edge of it.

- **Specific conductance.** The inner loop is split between the confluence count (13 folds) and the reach count (10), and between raw (14) and log (10). When the choice flips from fold to fold, so do the held-out predictions, and the extra options cost more than they gain. The two weightings are about equally good, so the tributary interpretation of Section 7 holds with either, but neither is better.
- **pH** moves to the intermediate weightings and chooses $\lambda = 10^{-6}$ in 17 of 19 folds. That is

effectively $\lambda \to 0$: no smoothing among training stations, with the held-out station predicted by interpolation along the network. The +0.008 is noise.

- **Water temperature** chooses $\lambda = 3000$, the new top of the grid, in 22 of 24 folds. Smoothing that strong makes the graph term nearly constant, which confirms the ablation result that water temperature needs no graph term.
- **Turbidity** never chooses the log target. This does not show that a log target cannot help. Raw scores the best of several hundred settings, while log is scored with a single setting tuned in log space, so the comparison favours raw.
- **DO** mostly chooses the confluence count, with no material change in R^2.

Decision. The six-weighting model of Tables 13 and 14 remains the main model and the script's default. The variants add computation and less stable choices, and do not improve any target. Choosing the simpler model after seeing these results is itself a post-hoc decision; the case for it is parsimony, not the 0.034. The variants stay available with `BENCHMARK_GRAPH_EXTENDED=1`.

11 Discharge as a predictor

Discharge is measured at many of the water-quality stations, and dilution and sediment transport both depend on flow. To test it, the hourly discharge of each station is mapped to its node (median over the node's stations, as for the water-quality variables) and enters as the anomaly

$$q_{\text{anom}} = \log Q - \overline{\log Q}_{\text{station}}.$$

Raw $\log Q$ mostly measures river size, which log upstream area already covers. The anomaly measures high or low flow relative to that station's own record, and uses only discharge, never water-quality values. Rows without discharge are dropped, which changes the task to "predict water quality at a flow-gauged station with no water-quality sensor". Two benchmark runs use exactly the same rows and outer folds:

- **baseline** (`BENCHMARK_DISCHARGE=baseline`): the inner-loop feature candidates of Section 4.2, without discharge;
- **with Q** (`BENCHMARK_DISCHARGE=on`): every candidate is also offered with q_{anom} added ("+ Q"), and the inner loop chooses.

The graph model is the six-weighting model of Table 14. Gradient boosting and random forest are still running, so this section makes no ranking claims against them.

The discharge rows are a harder task for the graph model. Before looking at discharge itself, note that the baseline graph model is much weaker on these rows than in Table 14: pH $0.117 \to -0.051$, specific conductance $0.385 \to 0.222$, DO $0.700 \to 0.554$ (Table 20). This is not because the held-out stations are harder: restricting the predictions of the main run to the discharge stations changes the graph model by 0.02 or less for pH, specific conductance and DO. It comes from training on fewer stations (pH 15 instead of 19, specific conductance 20 instead of 24) and fewer rows. The graph term interpolates between neighbouring stations, so it loses more from a sparser network than the linear models do. On these rows the graph model is no longer the best pH model (Spline GAM 0.078, Ridge 0.053), although it remains clearly best for specific conductance. The comparisons below are therefore only between the two discharge runs, not with the earlier tables.

Table 20: Subset diagnostic, R^2. "Main" is Table 14 (graph) and Table 9 (Ridge) on all its stations; "Main, Q stns." restricts those predictions to the discharge stations (indicative only, since those models were trained on all stations). "Q baseline" is trained and tested on the discharge rows only.

	Graph-Laplacian			Ridge		
Target	Main	Main, Q stns.	Q baseline	Main	Main, Q stns.	Q baseline
Water temperature	0.8634	0.8493	0.8290	0.8640	0.8502	0.8324
DO concentration	0.6996	0.6777	0.5536	0.7008	0.6716	0.6294
Spec. conductance	0.3847	0.3813	0.2222	−0.1096	−0.1334	−0.9910
pH	0.1173	0.1133	−0.0507	0.0029	0.0049	0.0530
Turbidity	0.1043	0.1811	0.1679	0.1362	0.1457	0.0996

Table 21: Effect of offering discharge: R^2 baseline → with Q on the same rows and folds, and the number of outer folds in which the inner loop chose a "+ Q" feature set. Bold: gains whose 95% station-bootstrap interval excludes zero and the change is not driven by one or two stations (see text).

Target (stations)	Graph-Laplacian	OLS	Ridge	Elastic Net	Spline GAM
Water temp. (20)	0.829 → 0.827	0.829 → 0.822	0.832 → 0.830	0.830 → 0.828	0.781 → 0.766
Q chosen	1/20	2/20	1/20	2/20	4/20
DO (14)	0.554 → 0.512	0.582 → 0.570	0.629 → 0.614	0.605 → 0.595	0.441 → 0.461
Q chosen	4/14	2/14	3/14	5/14	13/14
Spec. cond. (20)	0.222 → 0.259	**−1.338 → −1.282**	**−0.991 → −0.960**	**−1.023 → −0.983**	−0.172 → −0.175
Q chosen	18/20	17/20	17/20	17/20	1/20
pH (15)	−0.051 → −0.108	0.048 → 0.102	**0.053 → 0.132**	**−0.068 → 0.104**	**0.078 → 0.161**
Q chosen	15/15	15/15	15/15	15/15	14/15
Turbidity (14)	0.168 → 0.255	0.099 → 0.018	0.100 → 0.105	0.100 → 0.085	0.159 → 0.118
Q chosen	14/14	12/14	14/14	13/14	14/14

Results. Table 21 gives the paired changes. The intervals resample stations with the predictions held fixed, so they leave out the variability of the inner-loop choices and are, if anything, too narrow. Where most stations have exactly the same predictions in both runs (the inner loop chose the same set), an interval ending at 0.000 is an artifact of those ties and does not count as evidence.

- **pH.** Discharge helps the conventional models: Ridge +0.079 (interval 0.003 to 0.142), Elastic Net +0.172 (0.043 to 0.268), Spline GAM +0.083 (0.026 to 0.137), each better at 9–12 of 15 stations. The graph model chose "+ Q" in all 15 folds but lost 0.057 (interval −0.38 to 0.16). Ten stations improve; the loss comes from two stations whose error grows 2.3× and 3.6×. These are not short records. With discharge, the best pH model on these rows is Spline GAM (0.161).

- **Specific conductance.** Discharge helps every linear model by 0.03–0.06, with 13–14 of 20 stations better, consistent with dilution at high flow. Those models still have $R^2 < -0.9$, so this does not make them useful. The graph model gains 0.037, but its interval (−0.012 to 0.082) crosses zero and one station gives 54% of the gain. The graph model remains far ahead (0.259 vs at most −0.175).

- **Turbidity.** The graph model's +0.087 is not a finding. Its interval is −0.16 to 0.14, 7 stations improve and 7 get worse, one station gives 60% of the gain, and the gains are at the stations with the largest errors while several low-error stations get 2–3× worse. With row-pooled instead of equal-station weights the change is −0.039. The other models are unchanged or

worse (OLS −0.082, interval −0.52 to −0.01). Allowing a log target as well (graph-only, `BENCHMARK_GRAPH_LOG_TARGETS=Turbidity`) gives 0.168 → 0.236, but log was chosen in only 1 of 14 folds, so this is essentially the same comparison.

- **Water temperature and DO.** The inner loop rarely chooses discharge, except Spline GAM for DO, and the changes are small and mostly negative. The graph model's DO loss (−0.042) comes from the four folds that chose discharge, mostly one station.

Selection is not benefit. The inner loop choosing "+ Q" in most folds does not mean it helps the held-out station: the graph model chose it in 15/15 folds for pH and 14/14 for turbidity, and lost or showed no reliable gain. Doubling the candidates gives the inner loop more chances to choose a set that fits the training stations by chance.

Decision. Discharge stays opt-in (`BENCHMARK_DISCHARGE`). The only reasonably supported gains are for the conventional models on pH and, without practical value, for the linear models on specific conductance. It does not reliably help the graph model on any target, and requiring it removes stations, which costs the graph model more than discharge gives back.

11.1 Averaging the top graph settings

With `BENCHMARK_GRAPH_AVERAGE_TOP=3`, the graph model takes the best (λ, λ_β) within each (feature set, weighting) pair, ranks the pairs by inner-LOO MSE, and averages the predictions of the top three with equal weights. This targets the unstable fold-to-fold choices of Section 10. On the default model it changes no target by more than 0.006 (specific conductance 0.3847 → 0.3825, pH 0.1173 → 0.1232, DO 0.6996 → 0.7011, water temperature 0.8634 → 0.8630, turbidity 0.1043 → 0.1037). On the extended model of Table 19 it recovers 0.010 of specific conductance (0.3510 → 0.3610, still below the default's 0.3847) and raises pH by 0.006 (0.1248 → 0.1310), both within noise. For specific conductance and pH, the top three usually share the weighting and λ and differ only in feature set (in 92% and 84% of folds for the default model), so their predictions are nearly the same. Averaging stays off by default.

12 Five further graph-model ideas

Five changes were implemented as opt-in options of the benchmark and run graph-only on the rows and folds of Table 14. The target each one was meant to help was written down before the runs. With about 25 comparisons, changes on the other targets are treated as noise unless they are large.

1. **Dams** (`distance_dams`, `unweighted_dams`). Each large storage dam adds a resistance of 10× the median reach resistance to its reach, in series. 8 of 13 hand-listed dams lie on the network; Glen Canyon separates 22 from 3 stations, Hoover, Davis and Parker isolate one station, Flaming Gorge separates 1 from 7. The dams change the geometry (effective-resistance correlation with the dam-free matrix 0.92 for distance, 0.69 for unweighted). *Meant for* water temperature and turbidity.
2. **Flow-directed covariance** (`tailup_100`, `_300`, `_1000`). The exponential tail-up model of Ver Hoef et al. (2006): stations are correlated only along the flow path, $C_{ij} = \sqrt{A_i/A_j}\, e^{-d_{ij}/r}$ with upstream areas A, river distance d and range r in km. Its inverse replaces L_{eff}, without the sum-to-zero constraint. *Meant for* specific conductance and pH.

3. **REML** (`BENCHMARK_GRAPH_SELECT=reml`). Weighting and λ are chosen by restricted likelihood of the station-mean residuals (static MERIT predictors as fixed effects), feature set and λ_β still by inner LOO. *Meant for* more stable choices on every target.

4. **Network-varying seasonal cycle** (`BENCHMARK_GRAPH_VARYING=annual`). Station-specific coefficients on the first annual harmonic, smoothed over the graph with their own λ_v. *Meant for* water temperature, DO and turbidity.

5. **Graph + gradient boosting** (model `Graph-Laplacian + GB`). Gradient boosting fitted to the graph model's out-of-station residuals, with time-varying predictors only and one setting fixed in advance. *Meant for* turbidity and DO.

The solvers were checked against brute-force penalised least squares (agreement 10^{-13}), and the default model still reproduces Table 14 exactly.

Table 22: Change in R^2 against the main graph model (Table 14), on the same rows and folds, with the station-bootstrap 95% interval. †: a target the idea was meant to help. The last row compares the hybrid with gradient boosting of Table 14.

Idea	Water temp.	DO	Spec. cond.	pH	Turbidity
Dams	$-0.001^\dagger$ (-0.002, 0.000)	$+0.014$ (0.002, 0.032)	0.000	$+0.006$ (0.000, 0.015)	$+0.002^\dagger$ (-0.001, 0.002)
Tail-up	0.000 (-0.001, 0.001)	-0.007 (-0.034, 0.016)	$-0.227^\dagger$ (-0.380, 0.000)	$+0.004^\dagger$ (0.000, 0.016)	$+0.011$ (0.001, 0.055)
REML	$0.000^\dagger$ (-0.001, 0.000)	$+0.015^\dagger$ (-0.001, 0.037)	$-0.153^\dagger$ (-0.510, -0.020)	$-0.115^\dagger$ (-0.208, -0.032)	$-0.006^\dagger$ (-0.044, 0.000)
Varying	$-0.022^\dagger$ (-0.053, 0.001)	$-0.015^\dagger$ (-0.052, 0.009)	-0.005 (-0.031, 0.037)	-0.002 (-0.008, 0.002)	$+0.088^\dagger$ (0.013, 0.479)
Graph + GB	-0.018 (-0.048, 0.006)	$-0.009^\dagger$ (-0.054, 0.021)	-0.015 (-0.115, 0.024)	$+0.051$ (-0.158, 0.273)	$+0.244^\dagger$ (-0.024, 0.341)
Graph + GB, R^2	0.845	0.690	0.369	0.169	0.349
vs gradient boosting	$+0.013$	-0.043 (-0.074, -0.010)	$+0.463$	$+0.340$	-0.042 (-0.479, 0.064)

Results (Table 22).

- **Dams** do nothing for their intended targets: water temperature chose a dam weighting in 1 of 24 folds and turbidity in 4 of 17. DO chose one in 14 of 17 folds and gained 0.014 (interval 0.002 to 0.032, 9 stations better, 5 worse), but DO was not an intended target, so this is a hypothesis for another dataset, not a result.

- **Tail-up** was chosen often (water temperature 24/24, turbidity 16/17, DO 13/17) but changed no intended target for the better. The specific conductance loss of 0.227 comes from a single fold: at the highest-conductance station (96751415) the inner loop switched to `tailup_1000` and another feature set; every other fold is unchanged. It shows how much one station can move the specific conductance R^2, rather than a systematic effect.

- **REML makes things worse, and is not more stable.** It loses 0.153 for specific conductance and 0.115 for pH, with both intervals excluding zero, and the number of distinct choices across folds goes up (specific conductance: feature sets $3 \to 5$, (weighting, λ) pairs $2 \to 4$). For pH it chooses $\lambda = 100$ in 11 of 19 folds, where the inner LOO chose 10^{-4} in 16 of 19: the likelihood treats the station offsets as noise, while for predicting a new station they are the useful signal. Prediction-based selection is the right criterion here.

- **Varying seasonal cycle.** The inner loop chose the varying sets for water temperature (23 of 24 folds), turbidity (17/17), specific conductance (24/24) and DO (14/17), but rarely for pH (2/19). Of the three intended targets, water temperature (-0.022) and DO (-0.015) get slightly worse. Turbidity gains 0.088 (interval 0.013 to 0.48, 14 stations better, 3 worse), which fits the idea that the size and timing of the seasonal sediment peak differ along the network.

One station gives 69% of the gain, however, and the hybrid below gains far more. Specific conductance and pH are unchanged.

- **Graph + gradient boosting** is the only idea with a large effect. Turbidity rises from 0.104 to 0.349 (12 stations better, 5 worse), close to gradient boosting's 0.391, but the interval (-0.02 to 0.34) crosses zero and one station gives 64% of the gain. The hybrid keeps the graph model's lead for specific conductance (0.369 vs -0.094) and pH (0.169 vs -0.171), costs 0.018 for water temperature, and is 0.043 below gradient boosting for DO. It is the only model within 0.05 of the best model on every one of the five targets.

Decision. None of the four changes to the graph model itself is adopted; all stay opt-in. The varying seasonal cycle is a candidate for turbidity only. Graph + gradient boosting is worth confirming as a single model for all five targets, since its turbidity gain is large but rests on 17 stations and partly on one of them.

13 Other evaluation splits

Every result so far uses leave-one-station-out (LOO). Six other outer splits were run, all as opt-in options of the benchmark. In every one of them the hyperparameters and features are still chosen by inner leave-one-station-out on the training rows only. The splits answer different questions, so their R^2 values should be compared down a column, not across columns.

1. **Held-out time period** (`BENCHMARK_TIME_SPLIT=2026-01-01`). The cutoff was fixed in advance from the coverage of the records. A station is tested on its rows after the cutoff, using a model trained on the other stations before the cutoff. Each station needs at least 720 hours on its side of the cutoff. This leaves 16 test stations for water temperature and specific conductance, 12 for DO, 11 for pH and 8 for turbidity. Most records start in October 2025, so only 5–7 training stations cover each test month, and the test is partly a seasonal extrapolation (autumn and early winter to the rest of the year).

2. **One 80/20 station split** (`station_single`). 20% of the stations, chosen at random, are tested together. Unlike LOO, a test station's neighbours can be missing from training as well. Only 3–5 test stations per target.

3. **Twenty 80/20 station splits** (`station_repeated`). Twenty random draws of 3–5 test stations each. The pooled R^2 is over all draws: a station tested in several draws counts once, with all its test rows. Twenty draws do not reach every station, so 16–24 stations are tested per target instead of 17–24.

4. **Leave one subbasin out** (`subbasin`). The stations are grouped by station-number prefix into five subbasins, fixed before the run: Colorado headwaters (0907–0910), Gunnison and Colorado–Dolores (0911–0919), Green (0920–0933), San Juan (0934–0937), and Lower Colorado and Verde (0938–0951). Each subbasin is tested with all its stations held out together, so no test station has a training neighbour in its own subbasin. The subbasins hold 1–7 stations per target.

5. **Last 20% of each station** (`time_within`). Every station is in training with its first 80% of rows and tested on the rest. The graph model then knows each test station's own offset, which the conventional models, without a station identifier, cannot use. This is short-term forecasting at monitored stations, not prediction at unmonitored ones.

6. **Random 20% of rows** (`random_rows`). Neighbouring hours of the same station fall on both sides, so this split leaks heavily. It is an upper bound only.

Table 23: Held-out time period: R^2 on the test stations after the cutoff (equal station weights). The lower block gives the all-time LOO models (Tables 14 and 22) on the same stations and rows, i.e. trained with the future in view. Best in each block in bold.

	Water temp.	DO	Spec. cond.	pH	Turbidity
Test stations	16	12	16	11	8
Trained before the cutoff					
Dummy	−0.057	−0.029	−0.096	−0.051	−3.17
Ridge	0.749	0.603	−0.910	**−0.022**	−4.56
Gradient boosting	0.760	**0.683**	−0.116	−0.796	**−2.25**
Graph-Laplacian	**0.774**	0.543	**0.089**	−0.136	−5.22
Graph + GB	0.729	0.546	0.059	−0.471	−6.51
All-time LOO model, same rows					
Gradient boosting	0.797	**0.654**	−0.194	−0.402	**−0.50**
Graph-Laplacian	**0.858**	0.642	**0.229**	**0.151**	−1.61
Graph + GB	0.852	0.626	0.219	0.020	−1.36

Table 24: Held-out time period: paired ΔR^2 on identical test rows, with the station-bootstrap 95% interval (predictions held fixed). "Better/worse" counts the stations where the first model has the lower/higher mean squared error.

		Graph − gradient boosting			Graph + GB − graph		
Target	Stations	ΔR^2	Interval	Better/worse	ΔR^2	Interval	Better/worse
Water temp.	16	+0.014	(−0.10, 0.11)	7/9	−0.045	(−0.13, 0.01)	6/10
DO	12	−0.140	(−0.35, −0.03)	4/8	+0.004	(−0.04, 0.04)	6/6
Spec. cond.	16	+0.205	(−0.29, 0.56)	10/6	−0.030	(−0.39, 0.03)	4/12
pH	11	+0.659	(0.28, 1.32)	9/2	−0.334	(−0.76, −0.09)	1/10
Turbidity	8	−2.97	(−52.7, −0.42)	2/6	−1.29	(−17.9, 0.02)	1/7

Held-out time period (Tables 23 and 24).

- **Not seeing the future costs the graph model most for pH and specific conductance.** On the same rows its R^2 falls from 0.151 to −0.136 (pH), from 0.229 to 0.089 (specific conductance), from 0.858 to 0.774 (water temperature) and from 0.642 to 0.543 (DO). Gradient boosting loses less for water temperature, and for DO it does slightly better than its all-time model.
- **The graph model still beats gradient boosting for pH** (+0.66, interval 0.28 to 1.32, 9 of 11 stations better), but it is no longer better than predicting the training mean (Dummy −0.051, Ridge −0.022). In this setting no model predicts pH.
- **For specific conductance** the graph model is still the only model above zero, but the paired intervals cross zero.
- **For DO, gradient boosting is clearly better** (0.683 vs 0.543, interval excludes zero), unlike LOO, where the two were within 0.033.

- **Graph + GB does not hold up.** It is never better than the graph model alone and clearly worse for pH (−0.33, interval −0.76 to −0.09, 10 of 11 stations worse). The boosted residual correction learned on the earlier months does not carry over to the later ones.
- **Turbidity cannot be assessed.** Every model, including the training mean, is far below zero: the post-cutoff season has much lower variance and a different level than the training months. With 8 test stations the paired differences can refute a model but not confirm one. They point against both graph models.

Table 25: R^2 under one random 80/20 station split (`station_single`): 20% of the stations tested together, one outer fold. The best model in each column is in bold.

Model	Water temp.	DO	Spec. cond.	pH	Turbidity
Test stations	5	3	5	4	3
Dummy	−0.005	−0.240	−0.204	−0.102	−0.059
OLS	0.904	0.867	**−0.166**	−0.064	0.114
Ridge	0.907	0.866	−0.260	−0.041	0.114
Elastic Net	0.906	0.856	−0.263	−0.048	0.114
Spline GAM	0.880	0.822	−0.344	−0.032	−0.042
Random forest	0.593	0.860	−0.989	−0.117	−0.023
Gradient boosting	0.707	**0.878**	−0.647	−0.067	0.020
Graph-Laplacian	**0.907**	0.867	−1.442	0.079	**0.185**
Graph + GB	0.863	0.810	−1.546	**0.139**	0.101

Table 26: R^2 on the last 20% of each station's rows (`time_within`), with the first 80% of every station in training. Best in bold.

Model	Water temp.	DO	Spec. cond.	pH	Turbidity
Test stations	24	17	24	19	17
Dummy	−0.293	−0.324	−0.007	−0.081	−0.119
OLS	0.728	0.689	−0.392	−0.170	0.051
Ridge	0.729	0.693	−0.234	−0.084	0.045
Elastic Net	0.729	0.714	−0.280	**0.006**	0.046
Spline GAM	0.734	0.705	−0.079	−0.060	0.107
Random forest	0.788	0.727	−0.147	−0.268	**0.303**
Gradient boosting	**0.889**	**0.827**	−0.096	−0.078	0.239
Graph-Laplacian	0.728	0.728	**0.733**	−0.193	0.048
Graph + GB	0.735	0.711	0.667	−0.191	0.278

Single-fold splits (Tables 25–27; LOO values in Tables 14 and 22). Each has one outer fold, so no intervals are given. The 80/20 station split has only 3–5 test stations.

- **80/20 station split.** With whole groups of stations held out, the graph model ties for best on water temperature, is 0.011 behind gradient boosting for DO and is best for pH (with the hybrid) and turbidity. Specific conductance collapses to −1.44, the worst of all models. Two of the five test stations (96496279 and 203177183) account for this: the graph model's mean squared error there is 2–4× Ridge's, while at the other three it is similar or much better. The 20-draw run below shows that this was largely an unlucky draw.

Table 27: R^2 on a random 20% of rows (`random_rows`). Leaky: neighbouring hours of the same station fall in training and test. Upper bound only. Best in bold.

Model	Water temp.	DO	Spec. cond.	pH	Turbidity
Test stations	24	17	24	19	17
Dummy	0.000	0.000	0.000	0.000	−0.005
OLS	0.878	0.780	0.074	0.106	0.089
Ridge	0.878	0.780	0.072	0.096	0.089
Elastic Net	0.878	0.745	0.071	0.091	0.089
Spline GAM	0.888	0.775	0.266	0.397	0.236
Random forest	**0.995**	**0.987**	0.380	0.747	0.441
Gradient boosting	0.979	0.958	0.398	**0.856**	**0.644**
Graph-Laplacian	0.879	0.796	0.774	0.347	0.108
Graph + GB	0.930	0.883	**0.781**	0.619	0.442

- **Last 20% of each station.** With each station's own history available, gradient boosting is best for water temperature and DO, and random forest and the hybrid for turbidity. The graph model's specific conductance R^2 jumps to 0.73 because it knows each station's offset, which is most of the specific conductance signal. No model predicts the last 20% of pH (best 0.006).
- **Random rows** is leaky, as expected: random forest and gradient boosting reach 0.96–0.99 for water temperature and DO and 0.86 for pH, because they interpolate between neighbouring hours of the same station. These values say nothing about prediction at new stations or new times.

Table 28: Twenty random 80/20 station splits (`station_repeated`). Upper block: pooled R^2 over all draws (equal station weights). Lower block: mean ± standard deviation of the per-draw R^2 (3–5 test stations per draw). Best pooled value in bold. The Spline GAM pH value is dominated by one draw ($R^2 = -8{,}881$), in which its splines extrapolate on an out-of-range station.

Model	Water temp.	DO	Spec. cond.	pH	Turbidity
Test stations	24	16	24	18	16
Dummy	−0.030	−0.027	−0.075	−0.039	−0.030
OLS	0.859	0.683	−0.113	−0.009	0.144
Ridge	**0.861**	0.688	−0.167	0.001	0.140
Elastic Net	0.859	0.702	−0.164	0.010	0.141
Spline GAM	0.805	0.701	−0.060	−513	0.080
Random forest	0.745	0.732	−0.499	−0.299	0.121
Gradient boosting	0.826	**0.746**	−0.161	−0.102	0.364
Graph-Laplacian	0.859	0.702	**0.258**	0.047	0.181
Graph + GB	0.834	0.693	0.239	**0.054**	**0.421**
Per-draw R^2, mean ± s.d. over 20 draws					
Ridge	0.86 ± 0.06	0.68 ± 0.23	-0.47 ± 0.90	-0.05 ± 0.18	-0.45 ± 1.67
Gradient boosting	0.84 ± 0.13	0.74 ± 0.20	-0.68 ± 1.01	-0.26 ± 0.36	-0.12 ± 1.16
Graph-Laplacian	0.86 ± 0.06	0.71 ± 0.20	-0.02 ± 0.75	0.00 ± 0.19	-0.24 ± 1.16

Twenty 80/20 station splits (Table 28; LOO in Tables 14 and 22).

- **The LOO ranking survives.** Water temperature and DO are within about 0.01 of LOO for every linear and graph model. The graph model is still the only model clearly above zero for specific conductance (0.258; every non-graph model is below zero) and is best, with the hybrid, for pH.
- **The graph model loses most for specific conductance** (0.385 → 0.258). This fits its reliance on nearby stations: a draw that removes three stations often removes a test station's neighbours too. The subbasin split below tests this directly.
- **Single draws are very noisy.** The graph model's specific conductance R^2 ranges from −1.95 to 0.76 across draws (s.d. 0.75). The single 80/20 split (Table 25, −1.44) was at the bad end of this range, so a single 80/20 split should not be used to compare models here.

Table 29: Leave one subbasin out: pooled R^2 over all stations (equal station weights). Best in bold.

Model	Water temp.	DO	Spec. cond.	pH	Turbidity
Test stations	24	17	24	19	17
Dummy	−0.030	−0.026	**−0.135**	−0.048	−0.104
OLS	**0.852**	0.615	−1.440	−0.024	0.003
Ridge	0.851	0.624	−1.179	**−0.011**	−0.003
Elastic Net	0.850	0.643	−1.159	−0.020	−0.022
Spline GAM	0.760	0.654	−0.293	−0.143	−0.028
Random forest	0.699	0.675	−0.269	−0.487	0.122
Gradient boosting	0.763	**0.709**	−0.144	−0.392	0.028
Graph-Laplacian	0.851	0.623	−0.241	−0.119	0.015
Graph + GB	0.837	0.617	−0.244	−0.170	**0.171**

Table 30: R^2 within each subbasin, under LOO (only the station itself held out, so the rest of its subbasin is in training) and with the whole subbasin held out. Both are computed over the same stations and rows, with equal station weights. Within one subbasin R^2 is measured against that subbasin's own mean, so it is lower than the pooled values.

		Ridge		Gradient boosting		Graph-Laplacian	
Subbasin	Stations	LOO	Subbasin out	LOO	Subbasin out	LOO	Subbasin out
Specific conductance							
Colorado headwaters	7	−0.17	−0.25	−0.19	−0.14	0.52	−0.05
Green	6	−0.88	−1.34	−1.98	−2.54	0.37	−5.14
Gunnison + Colorado-Dolores	5	−0.30	−0.38	−0.23	−0.27	0.11	−0.20
Lower Colorado + Verde	3	−1.03	−23.2	−0.72	−0.58	−0.52	0.15
San Juan	3	−6.99	−10.0	−5.14	−6.41	−0.79	−9.79
pH							
Colorado headwaters	6	−0.00	0.02	−0.07	−0.21	0.16	−0.22
Green	2	−0.57	−0.66	−1.18	−1.37	−0.06	−0.79
Gunnison + Colorado-Dolores	4	−0.08	−0.25	−0.08	−0.35	0.15	0.12
Lower Colorado + Verde	3	−0.40	−0.40	−2.70	−2.25	−0.27	−0.13
San Juan	4	−0.05	−0.02	0.30	−0.29	−0.08	−0.38

Leave one subbasin out (Tables 29 and 30).

- **Water temperature transfers between subbasins.** The linear and graph models lose only about 0.01 against LOO (0.863 → 0.851). Water temperature follows climate and the annual cycle, which the subbasins share. DO loses about 0.08, and gradient boosting is best (0.709).
- **The graph model's specific conductance advantage does not transfer.** Its pooled R^2 falls from 0.385 (LOO) to −0.241, below the training mean (−0.135). Table 30 shows where the LOO skill comes from. Under LOO the graph model is positive within three subbasins (headwaters 0.52, Green 0.37, Gunnison 0.11), while Ridge and gradient boosting are negative in every subbasin. With the subbasin held out the graph model drops in four of five, most in the Green (0.37 → −5.14) and San Juan (−0.79 → −9.79). Ridge and gradient boosting change much less, because they never used the other stations of the subbasin. The graph model's specific conductance skill is therefore interpolation from stations in the same subbasin. It does not carry a salinity level into a subbasin with no training station.
- **pH behaves the same way, on a smaller scale.** Under LOO the graph model is the best of the three models within four of five subbasins. With the subbasin held out it drops in four of five, and no model is above zero in the pooled R^2 (best Ridge, −0.011).
- **The graph model degrades less than the linear models** for specific conductance (−0.24 vs −1.2 to −1.4; Ridge reaches −23 in the Lower Colorado), so its smoothing still guards against extreme errors.
- **Turbidity:** only the hybrid (0.171) and random forest (0.122) stay clearly above zero.
- Two subbasins hold a single station for some targets (San Juan for DO, Lower Colorado for turbidity). There the two splits are identical. The turbidity R^2 at that one station is below -10^4 for every model, because its series is nearly constant, so it is left out of Table 30 and of this discussion.

Which subbasins the LOO models predict well (Table 31). This uses the ordinary LOO predictions, grouped by subbasin.

- **The graph model's specific conductance and pH skill sits in the three upper subbasins.** For specific conductance it reaches 0.52 (headwaters), 0.37 (Green) and 0.11 (Gunnison), where every non-graph model except the Spline GAM in the headwaters is negative. For pH it is best or near best in the headwaters (0.16) and Gunnison (0.15).
- **In the San Juan, specific conductance is poorly predicted by every model**, but the graph model (−0.79) is far better than the rest (−4.7 to −7.0). This subbasin's stations differ in level more than any predictor explains.
- **Water temperature and DO are predicted well in four of five subbasins** (0.72–0.95), with gradient boosting or random forest best for DO in all four (0.01–0.18 above the graph model, the largest gap at the single San Juan station). Gradient boosting is also the only model clearly positive for turbidity, in the Gunnison (0.38) and San Juan (0.35).
- **The Lower Colorado and Verde fail for every target and model**, including water temperature (best 0.20). All three stations are tailwaters of large dams: 09380000 (Lees Ferry, below Glen Canyon Dam), 09510000 (Verde River below Bartlett Dam) and 09429490 (above Imperial Dam, below the Hoover–Davis–Parker chain). Their water temperature varies little within the record (standard deviation 2.4–4.4 °C, median 2.5, against medians of 5.7–6.5 °C in

Table 31: Leave-one-station-out R^2 within each subbasin, for every target: the benchmark predictions of Tables 14 and 22, grouped by subbasin. Each station is predicted with only itself held out. R^2 is computed over the subbasin's stations (equal station weights) against the subbasin's own mean. The best model other than Dummy is in bold; ties at two decimals are broken by the unrounded value. The single Lower Colorado turbidity station is omitted (near-constant series, $R^2 < -10^4$ for every model).

Subbasin	Stations	Dummy	Ridge	Spline GAM	Random forest	Gradient boosting	Graph-Laplacian	Graph + GB
Water temperature								
Colorado headwaters	7	−0.05	0.94	0.91	0.92	**0.95**	0.94	0.94
Gunnison + Colorado-Dolores	5	−0.14	0.88	0.88	0.93	**0.95**	0.88	0.91
Green	6	−0.02	0.86	0.83	0.72	**0.86**	0.86	0.84
San Juan	3	−0.11	0.87	0.88	**0.89**	0.87	0.87	0.81
Lower Colorado + Verde	3	−0.76	**0.20**	0.05	−1.15	−0.65	0.20	−0.01
DO								
Colorado headwaters	7	−0.04	0.76	0.75	**0.78**	0.77	0.75	0.74
Gunnison + Colorado-Dolores	4	−0.13	0.82	0.81	0.85	**0.86**	0.83	0.82
Green	2	−0.01	0.81	0.91	0.95	**0.95**	0.86	0.85
San Juan	1	−0.05	0.89	0.90	0.92	**0.94**	0.76	0.84
Lower Colorado + Verde	3	−0.26	−0.41	−0.43	−0.98	−0.45	**−0.37**	−0.45
Specific conductance								
Colorado headwaters	7	−0.10	−0.17	0.27	−0.21	−0.19	0.52	**0.55**
Gunnison + Colorado-Dolores	5	−0.34	−0.30	−0.29	−0.25	−0.23	0.11	**0.11**
Green	6	−0.92	−0.88	−1.56	−2.68	−1.98	**0.37**	0.21
San Juan	3	−6.26	−6.99	−4.73	−5.20	−5.14	**−0.79**	−1.35
Lower Colorado + Verde	3	−0.09	−1.03	**−0.11**	−1.35	−0.72	−0.52	−0.70
pH								
Colorado headwaters	6	−0.05	−0.00	0.06	−0.14	−0.07	**0.16**	0.10
Gunnison + Colorado-Dolores	4	−0.27	−0.08	−0.08	0.14	−0.08	0.15	**0.19**
Green	2	−0.61	−0.57	−0.21	−1.14	−1.18	**−0.06**	−0.41
San Juan	4	−0.05	−0.05	−0.13	−0.11	0.30	−0.08	**0.30**
Lower Colorado + Verde	3	−0.25	−0.40	−0.44	−2.62	−2.70	**−0.27**	−0.56
Turbidity								
Colorado headwaters	6	−0.72	−0.72	−1.01	−0.72	−0.38	−0.44	**−0.26**
Gunnison + Colorado-Dolores	4	−0.07	−0.05	−0.04	0.14	**0.38**	0.00	0.15
Green	2	−0.02	−0.07	−0.38	−0.37	−0.16	**−0.01**	−0.21
San Juan	4	−0.16	0.07	0.03	0.01	**0.35**	−0.00	0.32

the other subbasins; the record lengths differ, so this comparison is indicative). The dams release deep reservoir water with a muted, delayed annual cycle. A seasonal model fitted mostly on unregulated stations then predicts a cycle that is not there. The small within-subbasin variance also makes R^2 harsh here, so moderate errors give a low value.

- The dam variant of Section 12 only changes the graph geometry, so it cannot change the seasonal terms that fail here. A tailwater indicator, or a seasonal amplitude allowed to differ below major dams, would target this directly. It was not tried, and with only three such stations it would need care to avoid tuning to them.

Summary. The graph model's main advantage (specific conductance and pH at unmonitored stations) holds under the held-out time period against gradient boosting, but there it no longer beats the training mean for pH. It survives twenty 80/20 station draws, at a lower level for specific conductance (0.258); the single 80/20 draw was an unlucky one. It does not survive leave one subbasin out: the specific conductance skill is interpolation between stations of the same subbasin, so the model should be used where its subbasin has monitored stations, and not to predict a basin with no stations. Water temperature transfers well under every split. The Graph + GB hybrid, promising under LOO (Section 12), does not carry over to a new time period and should not be adopted on the current evidence.

14 Upstream-watershed and neighbour-reading features

Two further predictors were added as opt-in options and tested under LOO with the five fast models (Dummy, OLS, Ridge, Elastic Net, Spline GAM) and the graph model. Each is offered to the inner loop as an extra candidate, so it is used only where the inner loop chooses it. The comparisons are paired against the runs of Table 14 on identical rows (`compare_runs.py`), with the station-bootstrap interval of Table 21.

1. **Upstream-watershed attributes** (`BENCHMARK_FS_STATIC=...,upstream`). A fourth static candidate: local slope plus three EPA StreamCat metrics accumulated over each station's whole upstream watershed, chosen in advance for specific conductance: mean lithological sulfur (gypsum and evaporites), mean lithological calcium oxide (carbonates) and row-crop land in 2019 (irrigation return flow). Stations are matched to their NHDPlus reach through the USGS NLDI. One station (09217000), whose reach has no StreamCat values, uses the first reach downstream.
2. **Neighbour reading** (`BENCHMARK_NEIGHBOUR=on`). The mean anomaly (z-score against each station's own training mean and standard deviation) of the target at the same hour at the three nearest training stations, nearest by effective resistance on the distance-weighted Kron Laplacian; 0 when none of them reports. At least 89% of rows have a concurrent neighbour. Every feature candidate is offered with and without it. The held-out station's own data never enters its feature. The inner loop reuses the features of the outer fold, so its choice is mildly optimistic.

Table 32: Upstream-watershed attributes: paired ΔR^2 against the same model without them (LOO, identical rows), with the station-bootstrap 95% interval. Better/worse counts stations with lower/higher mean squared error. Water temperature and turbidity are unchanged for every model (the set was never chosen). Intervals that exclude zero are in bold.

Target	Model	Without	With	ΔR^2	Interval	Better/worse
Spec. cond.	Graph-Laplacian	0.385	0.366	−0.019	(−0.12, 0.03)	9/11
	OLS	−0.099	−0.220	−0.121	(−0.20, 0.00)	0/1
	Ridge	−0.110	−0.217	−0.107	(−0.28, 0.00)	0/2
	Elastic Net	−0.109	−0.207	−0.098	(−0.33, 0.00)	0/2
	Spline GAM	0.022	−0.004	−0.026	(−0.04, 0.00)	0/1
DO	Graph-Laplacian	0.700	0.697	−0.002	(−0.02, 0.02)	2/2
	OLS	0.695	0.692	−0.003	(−0.03, 0.02)	1/1
	Ridge	0.701	0.695	−0.006	(−0.03, 0.02)	1/2
	Elastic Net	0.705	0.687	−0.018	(−0.04, 0.00)	4/11
	Spline GAM	0.708	0.688	**−0.020**	(−0.05, −0.00)	0/3
pH	Spline GAM	0.012	−0.050	−0.061	(−0.19, 0.00)	0/1

Upstream-watershed attributes (Table 32).

- **No gain for any model or target.** For specific conductance, the target they were chosen for, the graph model selected them in 20 of 24 outer folds, yet its R^2 fell slightly (−0.019, interval −0.12 to 0.03, 9 stations better and 11 worse). The linear models fell by 0.10–0.12 in the few folds where they chose them.

Table 33: Neighbour reading: paired ΔR^2 against the same model without it (LOO, identical rows), with the station-bootstrap 95% interval. Better/worse as in Table 32. Intervals that exclude zero are in bold.

Target	Model	Without	With	ΔR^2	Interval	Better/worse
Water temp.	Graph-Laplacian	0.863	0.874	+0.010	(−0.01, 0.03)	18/6
	OLS	0.863	0.875	+0.011	(−0.01, 0.03)	19/5
	Ridge	0.864	0.875	+0.011	(−0.01, 0.03)	19/5
	Elastic Net	0.863	0.875	+0.011	(−0.01, 0.03)	19/5
	Spline GAM	0.838	0.843	+0.004	(−0.02, 0.03)	13/5
DO	Graph-Laplacian	0.700	0.712	+0.012	(−0.01, 0.05)	12/5
	OLS	0.695	0.701	+0.006	(−0.02, 0.02)	14/3
	Ridge	0.701	0.705	+0.004	(−0.02, 0.03)	14/3
	Elastic Net	0.705	0.721	+0.016	(−0.00, 0.04)	13/4
	Spline GAM	0.708	0.699	−0.009	(−0.02, 0.00)	0/2
Spec. cond.	Graph-Laplacian	0.385	0.392	+0.007	(−0.03, 0.06)	14/10
	OLS	−0.099	−0.040	**+0.059**	(0.02, 0.20)	21/3
	Ridge	−0.110	−0.060	**+0.049**	(0.01, 0.17)	21/3
	Elastic Net	−0.109	−0.060	**+0.049**	(0.02, 0.16)	21/3
	Spline GAM	0.022	0.020	−0.002	(−0.01, 0.00)	0/1
pH	Graph-Laplacian	0.117	0.178	+0.061	(−0.16, 0.18)	17/2
	OLS	−0.014	0.130	**+0.143**	(0.06, 0.22)	15/4
	Ridge	0.003	0.121	**+0.118**	(0.06, 0.18)	14/5
	Elastic Net	0.023	0.143	**+0.121**	(0.04, 0.19)	14/5
	Spline GAM	0.012	0.153	**+0.141**	(0.05, 0.23)	13/6
Turbidity	Graph-Laplacian	0.104	0.088	−0.016	(−0.14, 0.03)	8/7
	OLS	0.136	0.125	−0.011	(−0.08, 0.00)	2/8
	Ridge	0.136	0.126	−0.010	(−0.08, 0.00)	3/8
	Elastic Net	0.136	0.126	−0.010	(−0.08, 0.00)	3/8
	Spline GAM	0.091	0.062	−0.029	(−0.06, 0.06)	10/7

- For the graph model, neighbouring stations on the network share most of their upstream geology, so the Laplacian smoothing probably already carries what these attributes add. For the linear models, three more station-level predictors over about 24 stations add another way to overfit.
- The attributes were chosen before any run, so this is a fair negative result. The option stays off by default.

Neighbour reading (Table 33).

- **It clearly helps pH.** Every conventional model gains 0.12–0.14 with intervals above zero, which brings them from about 0 to 0.12–0.15, level with the graph model without it. The graph model rises to 0.178, the best LOO pH value so far, with 17 of 19 stations better. Its interval crosses zero because one or two stations swing a lot.
- **Specific conductance:** the linear models gain about 0.05 (21 of 24 stations better) but stay below zero. The graph model barely changes (+0.007): its smoothing already uses the information that nearby stations carry.

- **Water temperature and DO:** consistent small gains of about 0.01 for most models, with intervals crossing zero.
- **Turbidity:** a small loss for every model, consistent with the weak correlation between a station's turbidity anomaly and its neighbours' (median 0.28, against 0.96 for water temperature).
- The inner loop chose it in every fold for every linear model and the graph model for water temperature, DO, specific conductance and pH.
- **It answers a different question.** The feature needs live readings from neighbouring stations at the same hour, so these results apply to filling in an unmonitored site while its neighbours report, not to prediction from map features alone. Both tasks are useful, but values with and without it should be reported separately. It stays off by default.

15 Longer training history

The benchmark rows of each station cover only part of its record (85% of the water-temperature test rows are from 2025–26). `download_history.py` downloads every benchmark station and variable from water year 2021 (fixed before any results) to the end of each series, and `build_history.py` builds extended tables that keep every current row unchanged and add 862,708 node-hours outside them (Table 34 gives the counts). On the hours where both exist, 93–97% of the values match the current data exactly. With `BENCHMARK_HISTORY=data/benchmark_data_history`, the added rows of the training stations enter training, while the held-out station is still tested on exactly its current rows and its added rows are never used, so the comparison with Table 14 is paired on identical test rows. The run covers the five fast models and the graph model.

Table 34: Node-hours of the current benchmark and those added by the longer history (water years 2021–2026).

Variable	Current	Stations	Added	Stations
Water temp.	151,801	25	862,174	25
DO	115,356	17	504,134	16
Spec. cond.	150,997	25	744,930	24
pH	122,122	19	510,860	18
Turbidity	90,843	17	391,022	15

Results (Tables 35 and 36).

- **The longer history helps no target for the graph model.** Water temperature and pH fall with intervals below zero; specific conductance and turbidity fall within noise; DO is unchanged. For the other models it helps only specific conductance (linear models +0.07 to +0.08, intervals above zero), where they stay below zero, and DO slightly (about +0.008, within noise). The graph model's lead for specific conductance shrinks from +0.36 to +0.28 over the next model (Spline GAM, 0.055).
- **Water temperature: a common cold offset.** With the history every model predicts the held-out stations about 1 °C too cold on average (mean signed station bias −0.92 to −0.98 °C for the graph and linear models, against −0.05 before). The within-station error variance

Table 35: Longer history: paired ΔR^2 against the same model trained on the current rows only (LOO, identical test rows), with the station-bootstrap 95% interval. Better/worse counts stations with lower/higher mean squared error. Intervals that exclude zero are in bold.

Target	Model	Current	History	ΔR^2	Interval	Better/worse
Water temp.	Graph-Laplacian	0.863	0.816	**−0.048**	(−0.11, −0.01)	11/13
	OLS	0.863	0.840	**−0.024**	(−0.04, −0.01)	10/14
	Ridge	0.864	0.840	**−0.024**	(−0.05, −0.01)	11/13
	Elastic Net	0.863	0.841	**−0.023**	(−0.04, −0.01)	10/14
	Spline GAM	0.838	0.745	−0.094	(−0.28, 0.01)	10/14
DO	Graph-Laplacian	0.700	0.699	−0.001	(−0.02, 0.02)	9/8
	OLS	0.695	0.702	+0.007	(−0.01, 0.02)	10/7
	Ridge	0.701	0.708	+0.008	(−0.00, 0.02)	11/6
	Elastic Net	0.705	0.712	+0.007	(−0.01, 0.03)	9/8
	Spline GAM	0.708	0.719	+0.011	(−0.00, 0.03)	13/4
Spec. cond.	Graph-Laplacian	0.385	0.334	−0.050	(−0.11, 0.06)	10/14
	OLS	−0.099	−0.022	**+0.077**	(0.01, 0.24)	15/9
	Ridge	−0.110	−0.032	**+0.077**	(0.02, 0.24)	15/9
	Elastic Net	−0.109	−0.039	**+0.070**	(0.01, 0.22)	15/9
	Spline GAM	0.022	0.055	+0.033	(−0.01, 0.19)	17/7
pH	Graph-Laplacian	0.117	−0.037	**−0.155**	(−0.29, −0.05)	8/11
	OLS	−0.014	−0.011	+0.003	(−0.07, 0.07)	12/7
	Ridge	0.003	0.013	+0.010	(−0.05, 0.08)	12/7
	Elastic Net	0.023	0.029	+0.006	(−0.03, 0.04)	8/11
	Spline GAM	0.012	0.009	−0.003	(−0.12, 0.10)	13/6
Turbidity	Graph-Laplacian	0.104	0.043	−0.061	(−0.42, 0.33)	8/9
	OLS	0.136	0.032	−0.104	(−0.26, 0.04)	9/8
	Ridge	0.136	0.031	−0.105	(−0.21, 0.05)	9/8
	Elastic Net	0.136	0.033	−0.104	(−0.20, 0.05)	9/8
	Spline GAM	0.091	0.085	−0.006	(−0.07, 0.31)	12/5

is unchanged (5.96 → 6.01); the loss is all in the station-level bias. The model has no year term, so the longer history moves its level to the 2021–26 average, while the test rows are mostly from 2025–26 and water year 2026 was 1.2 °C warmer than each station's usual value for the day of year (water year 2023 was 1.1 °C colder; the other years are within 0.1). In the current benchmark the training and test stations cover the same months, so this offset cancels. Subtracting a single common offset (a descriptive check, not a fair model) restores Ridge to 0.860 and the graph model to 0.840.

- The rest of the graph model's water-temperature loss is one fold: with 09429490 (above Imperial Dam) held out, the inner loop chose $\lambda_L = 0.03$ instead of 30–100 as in every other fold, and that station's bias went from +0.2 to −5.8 °C.
- **pH: the inner loop changes its choice.** On the current rows it chose $\lambda_L \approx 10^{-4}$ in nearly every fold; with the history it mostly chooses $\lambda_L = 3$ and $\lambda_\beta = 3$. The mean signed bias stays near zero, but the station biases spread (standard deviation 0.11 → 0.15), so the graph model loses the station-level information that gave it its lead. The conventional models do not change.
- **Turbidity:** the linear models lose about 0.10 through larger station-level bias. The added

Table 36: Longer history on the upper-basin stations: LOO R^2 current → history, scored on the stations of each subset. The models are those of Table 35, trained on all stations, so this is a breakdown of the same predictions, not an upper-basin retrain. The subsets were chosen after seeing the results and are descriptive only.

Subset	Target	Stations	Graph-Laplacian	Ridge	Spline GAM
Upper Basin (excl. Lower)	Water temp.	21	0.899 → 0.876	0.899 → 0.874	0.880 → 0.871
	DO	14	0.795 → 0.801	0.800 → 0.808	0.810 → 0.815
	Spec. cond.	21	**0.425 → 0.391**	−0.068 → 0.018	0.027 → 0.053
	pH	16	0.140 → 0.028	0.024 → 0.067	0.039 → 0.087
	Turbidity	16	0.103 → 0.105	0.144 → 0.066	0.090 → 0.083
Excl. San Juan and Lower	Water temp.	18	0.900 → 0.876	0.901 → 0.872	0.877 → 0.867
	Spec. cond.	18	**0.381 → 0.345**	−0.074 → −0.007	−0.003 → 0.023
	pH	12	0.216 → 0.114	0.016 → 0.067	0.082 → 0.142
Headwaters only	Water temp.	7	0.937 → 0.934	0.937 → 0.936	0.909 → 0.932
	Spec. cond.	7	**0.521 → 0.514**	−0.174 → 0.023	0.269 → 0.329
	pH	6	0.161 → 0.017	−0.003 → 0.014	0.056 → 0.026

years contain readings of 3,800–4,000 FNU that look like sensor maxima; whether they cause the loss was not tested.

- **The upper basin shows the same pattern** (Table 36). Without the Lower Colorado, water temperature still falls by about 0.025 for every model with a similar offset (−0.8 °C), so the loss is not caused by the dam tailwaters. In the headwaters alone the offset is only −0.2 °C and water temperature holds at about 0.935. The graph model's specific conductance falls by 0.01–0.04 but keeps a large lead, and its pH falls in every subset, where the Spline GAM overtakes it.
- **Next step.** The fair way to use the history is to add a water-year (or monthly) effect estimated from the training stations, which are observed at the same times as the held-out station, and to clip turbidity at the sensor maximum; an upper-basin-only retrain would then test whether the extra years help there. The option stays off by default.

16 Conclusions

1. **Kron reduction is exact for this model.** It matches the full 67,401-node Laplacian to rounding error for every weighting, and gives identical hyperparameters and R^2. It is about 150× faster, so the 26×26 Kron matrix should remain the default. Compressing the graph changes nothing either.
2. **The value of the graph term depends on the target.** It clearly helps for pH and, to a lesser extent, specific conductance. For water temperature, DO and turbidity, seasonal and MERIT predictors matter far more than the Laplacian.
3. **Report the fair numbers.** The fair R^2 values in Table 5 should replace earlier figures. In particular, specific conductance should be reported at about 0.14, not 0.22–0.27.
4. **Use the updated benchmark for model comparisons.** The graph model is now fitted inside `water_quality_model_benchmark.py` on the same data as the other models, and the

full rerun (Table 7) replaces the hand-entered graph rows of the Nov 5 table, in particular turbidity (0.0145 → 0.026).

5. **The graph model's advantage survives fair feature selection, but only for two targets.** With every model choosing its features inside the inner loop (Table 9), the graph model is still clearly best for specific conductance (0.207 vs at most 0.022) and pH (0.130 vs at most 0.023). It ties with the linear models for water temperature, is mid-pack for DO, and trails gradient boosting by 0.28 for turbidity. Inner-loop selection should be the default for future benchmark runs.

6. **Edge weights matter only on the compressed graph, and only for specific conductance.** On the original graph, removing weights changes almost nothing because the segments are nearly uniform in length. On the compressed graph, an unweighted Laplacian (one step per reach) is a different geometry. Chosen inside the inner loop, it raises the graph model's specific conductance R^2 from 0.207 to 0.385 and leaves the other targets within noise (Table 13).

7. **Final standing.** With fair feature selection for every model and the updated graph model (Table 14), the graph-Laplacian model is clearly the best choice for specific conductance (+0.36 over the next model) and pH (+0.10). It is no better than simpler models for water temperature and DO. For turbidity, gradient boosting is the model to use.

8. **Ablation confirms where the graph term matters.** It is the most valuable input for specific conductance (−0.45 when removed) and pH (−0.11), adds nothing for water temperature and DO, and hurts turbidity (Tables 15 and 16). Water temperature is driven by the annual cycle, DO by water temperature and the daily cycle, and turbidity by the annual cycle. Among the MERIT variables only slope consistently helps.

9. **With all chemistry available, pH and DO explain each other.** On the 14 stations that measure all five variables (Section 9.1), DO is the most valuable predictor of pH and pH of DO, consistent with photosynthesis driving both. The graph term then matters little for pH but remains essential for specific conductance (−0.42 when removed).

10. **Further variants do not help.** A confluence-count Laplacian, length-power weights, a log target and a wider λ grid, all chosen inside the inner loop, change no target beyond noise, and specific conductance falls by 0.034 (Table 19). With this few stations, extra options make the inner-loop choices less stable. The six-weighting model remains the main model.

11. **Discharge does not help the graph model, and requiring it hurts.** On the rows with discharge (Section 11), the discharge anomaly reliably helps only the conventional models for pH (Ridge +0.08, Spline GAM +0.08, Elastic Net +0.17) and the linear models for specific conductance, where they stay below zero (Table 21). For the graph model no change is reliable, and keeping only flow-gauged rows removes training stations, which lowers the graph model's pH R^2 from 0.117 to −0.051 before discharge is even used (Table 20). Averaging the top three graph settings changes nothing beyond noise. Both options stay off by default.

12. **Dams, flow-directed covariance, REML and a network-varying seasonal cycle do not improve the graph model** (Section 12), apart from a possible +0.09 for turbidity with the varying seasonal cycle. REML is clearly worse for specific conductance and pH, because it treats the station offsets as noise. Only graph + gradient boosting changes a target materially: turbidity 0.104 → 0.349, close to gradient boosting (0.391), while keeping the graph model's

lead for specific conductance and pH. The turbidity interval still crosses zero, so this hybrid should be confirmed before it is adopted.

13. **Under other splits the picture is mixed** (Section 13). On a held-out time period the graph model still beats gradient boosting for pH (+0.66, interval excludes zero) and is the only model above zero for specific conductance, but no model beats the training mean for pH, and gradient boosting is clearly better for DO. The Graph + GB hybrid is no better than the graph model alone on the held-out period and clearly worse for pH, so it is not adopted. Over twenty 80/20 station draws the LOO ranking holds, with the graph model's specific conductance lower (0.258); the single 80/20 draw (−1.44) was an unlucky one.

14. **The graph model interpolates within subbasins; it does not extrapolate to new ones.** With a whole subbasin held out, its specific conductance R^2 falls from 0.385 to −0.241, below the training mean. Under LOO it is positive within three subbasins while Ridge and gradient boosting are negative in all five (Table 30), so its skill comes from other stations of the same subbasin. Water temperature transfers between subbasins with almost no loss. Claims about the graph model should be limited to sites in subbasins that already have monitored stations. Under LOO its skill sits in the three upper subbasins (Table 31). Every model fails in the Lower Colorado and Verde, whose three stations are all below large dams, even for water temperature (best 0.20).

15. **Upstream-watershed geology and crop land do not help** (Table 32), although the inner loop chose them for the graph model in 20 of 24 specific conductance folds. The network smoothing already carries most of that information.

16. **Concurrent readings at neighbouring stations clearly help pH** (Table 33): +0.12 to +0.14 for every conventional model, with intervals above zero, and 0.117 → 0.178 for the graph model, the best LOO pH so far. They add about 0.05 to the linear models for specific conductance and nothing to the graph model. This is a different task (live neighbours required), so it should be reported separately from prediction from map features alone.

17. **A longer training history (water years 2021–2026) does not help as used** (Section 15). Water temperature falls for every model (−0.02 to −0.05, intervals below zero) because the model, with no year term, learns the multi-year level while the test rows come from a warmer recent period, and the graph model loses its pH lead (0.117 → −0.037) as the inner loop moves to heavier shrinkage. Only the linear models gain, for specific conductance (+0.08), and they stay below zero. The same holds on the upper-basin stations. A water-year effect is needed before the history can be judged.

Caveats. With only 14–24 stations, differences in R^2 below roughly 0.03–0.05 are within noise. Standard errors were computed only for the discharge comparison (Table 21), as station-bootstrap intervals that hold the inner-loop choices fixed. For pH and specific conductance the chosen features vary between folds, another sign that those gains are fragile. The daily×annual interaction terms and separate daily and annual K from the original water-temperature section were not among the notebook's candidates. In the original they changed predictions by at most 0.009 °C. In the benchmark, the "configured" candidate was designed using outer-LOO results, so the inner-selection values in Table 9 may still be slightly optimistic for every model, most likely for specific conductance. The two-stage search chooses each non-graph model's features at a single reference setting, which may disadvantage models whose best features depend strongly on their settings. The unweighted compressed Laplacian was only tried because an exploratory run (Table 12) looked good for specific

conductance, a mild second-order selection effect that the inner loop cannot remove. The gain (+0.18) is much larger than that effect plausibly explains, but it should be confirmed on another basin or a held-out time period. A graph-only server run reproduced the Table 13 values to 10^{-12}. The main ablation uses only each target's fixed chemistry predictors. The all-chemistry version (Section 9.1) is limited to 14 stations, so its results are noisy, its full models overfit, and its R^2 values are not comparable with the benchmark. The discharge runs are a different task (flow-gauged stations only, 14–20 stations), so their R^2 values are not comparable with Table 14, and their gradient boosting and random forest runs had not finished when this section was written. The neighbour-reading and upstream runs cover only the five fast models and the graph model, and so does the longer-history run. Gradient boosting with the neighbour reading had not finished when this section was written. In the neighbour run the inner loop reuses the outer fold's neighbour features, so its choices are mildly optimistic.

Reproducibility. All results come from `largest_uncompressed_laplacian_fixed.ipynb` and are saved in `results/laplacian_fixed/`. The standalone check script is `compare_kron_vs_full_laplacian.py`. The benchmark comparisons in Tables 7 and 8 come from a full server run of `water_quality_model_benchmark.py` (commit `33e913b`), saved in `results/benchmark_results/`. The graph rows can be rerun on their own with `BENCHMARK_MODELS=Graph-Laplacian`. Table 9 comes from a full server run of the same script at commit `b72b9b4`, with its defaults (inner selection, two-stage search, stride 1). Its results, including `feature_selection_counts.csv` with the feature set chosen in each outer fold, are written to `results/benchmark_results_inner_fs/`. To reproduce the fixed-feature tables, set `BENCHMARK_FEATURE_SELECTION=fixed`. Tables 11 and 12 come from `compare_edge_weights.py` (about a minute), saved in `results/edge_weight_matrix_checks.csv` and `results/edge_weight_r2.csv`. Table 13 and the graph row of Table 14 come from a graph-only server run of the benchmark at commit `c1b3dde` (`BENCHMARK_MODELS=Graph-Laplacian`), saved in `results/benchmark_results_inner_fs_graph/`. The script's current defaults reproduce this model exactly (six weightings, λ from 10^{-4} to 100, raw targets). Setting `BENCHMARK_GRAPH_WEIGHTINGS=distance_only,area,width,slope,area_width` and `BENCHMARK_GRAPH_LAMBDA_GRID=0.01,0.03,0.1,0.3,1,3,10,30,100` restores the earlier five-weighting graph model of Table 9. Table 19 comes from a graph-only server run at commit `f9f9f42`, whose defaults then included all four variants. The same model is now run with `BENCHMARK_GRAPH_EXTENDED=1`, and its results are saved in `results/benchmark_results_graph_improved/`. Tables 15 and 16 come from `ablation_study.py` (commit `56a3ccb`, under two minutes with `--jobs 6`), saved in `results/ablation/`. Tables 17 and 18 come from `ablation_study.py --all-chemistry` (commit `3249fe6`), saved in `results/ablation_all_chemistry/`. Server runs reproduced both to 10^{-9}. Tables 20 and 21 come from server runs at commit `095727e` with `BENCHMARK_MODELS=Graph-Laplacian,OLS,Ridge,Elastic Net,Spline GAM` and `BENCHMARK_DISCHARGE=baseline` or `on`, saved in `results/benchmark_results_discharge_baseline/` and `results/benchmark_results_discharge_on/`. The log-turbidity variants are in `results/q_baseline_logturb/` and `results/q_on_logturb/`. The paired comparison, bootstrap intervals and subset diagnostic come from `discharge_paired_analysis.py` (seconds), saved in `results/discharge_paired_comparison.csv` and `results/discharge_subset_diagnostic.csv`. The averaging results (Section 11.1) come from graph-only runs at the same commit with `BENCHMARK_GRAPH_AVERAGE_TOP=3`, without and with `BENCHMARK_GRAPH_EXTENDED=1`, saved in `results/avg3_default/` and `results/avg3_extended/`. Table 22 comes from server runs at commit `4949dbe`, saved in `results/idea_dams/`, `idea_tailup/`, `idea_reml/`, `idea_varying/` and `idea_graph_gb/`. The paired comparisons come from `compare_runs.py` and are saved as `results/idea_*_`

vs_*.csv. Tables 23 and 24 come from a server run at commit 220357a with BENCHMARK_TIME_SPLIT=2026-01-01 and BENCHMARK_MODELS=Dummy,Ridge,Graph-Laplacian,Graph-Laplacian + GB,Gradient Boosting, saved in results/benchmark_results_time_split/; the paired comparisons and the all-time reference come from time_split_analysis.py and are saved in the same folder as time_split_paired.csv and time_split_reference.csv. Tables 25–27 come from server runs at commit 8e1d542 with BENCHMARK_SPLIT=station_single, time_within or random_rows (nine models), saved in results/benchmark_results_split_*/, and from split_analysis.py, saved in results/split_comparison.csv. Table 28 comes from a server run at the same commit with BENCHMARK_SPLIT=station_repeated, saved in results/benchmark_results_split_station_repeated/; the per-draw statistics are in results/split_repeated_draws.csv. Tables 29–33 come from server runs at commit bf33e18: BENCHMARK_SPLIT=subbasin (nine models), BENCHMARK_FS_STATIC=none,local_slope,reduced,upstream and BENCHMARK_NEIGHBOUR=on (both with BENCHMARK_MODELS=Dummy,OLS,Ridge,Elastic Net,Spline GAM,Graph-Laplacian), saved in results/benchmark_results_split_subbasin/, results/benchmark_results_upstream/ and results/benchmark_results_neighbour/. The per-subbasin values, with the LOO R^2 over the same stations, are in results/split_subbasin_folds.csv (split_analysis.py). The upstream attributes come from fetch_upstream_attributes.py (public NLDI and StreamCat APIs, about a minute), saved in data/benchmark_data/upstream_attributes.csv. The paired comparisons come from compare_runs.py against results/benchmark_results_inner_fs_graph/ (graph model) and results/benchmark_results_inner_fs/ (other models), saved as results/upstream_vs_base_*.csv and results/neighbour_vs_base_*.csv. Tables 34–36 come from download_history.py (USGS Water Data API, server, a few hours), build_history.py (written to data/benchmark_data_history/, not tracked) and a server run with BENCHMARK_HISTORY=data/benchmark_data_history and BENCHMARK_MODELS=Dummy,OLS,Ridge,Elastic Net,Spline GAM,Graph-Laplacian, saved in results/benchmark_results_history/. The paired comparisons are results/history_vs_base_*.csv (compare_runs.py); the bias decomposition, the common-offset check, the regional subsets and the water-year anomalies come from history_analysis.py (seconds), saved in results/history_diagnostics.csv, results/history_vs_base_by_region.csv and results/history_wt_anomaly_by_wy.csv.